CODE-MM: Convex Deep Mangrove Mapping Algorithm Based on Optical Satellite Images

Chia-Hsiang Lin, *Member, IEEE*, Man-Chun Chu, *Student Member, IEEE*, and Po-Wei Tang, *Student Member, IEEE*

***Abstract*— Mangrove mapping (MM) is a critical satellite remote sensing technology since mangrove forests have a large capacity for carbon storage among the blue carbon ecosystems. However, we surprisingly found that benchmark MM methods are all index-based ones, completely ignoring the spatially neighboring information on the one hand and quite sensitive to the threshold setting on the other hand. Deep learning has been proven to be an effective solution for incorporating the desired spatial information, but the induced big data collection of MM is difficult and time-consuming, especially for ground-truth labeling; this would be the reason why benchmark methods are all index-based ones. To solve the dilemma, we introduce convex analysis into deep learning, thereby achieving small-data learning. The proposed algorithm is hence termed convex deep MM (CODE-MM), mainly developed for the Sentinel-2 satellite, which is the mainstream satellite for the MM mission, as it involves those key green/infrared bands for characterizing mangrove multispectral signatures. We also generalize our CODE-MM to test the hyperspectral satellite data, which should be the trend for various classification missions in the future due to its strong material identifiability. Simply speaking, CODE-MM first infers a rough mangrove signature for designing a Siamese deep regularizer, which is then plugged into a convex criterion customized for the mapping task. We implement the convex criterion by deriving closed-form solutions for all the algorithmic steps, ensuring computational efficiency. Extensive experiments demonstrate that CODE-MM is insensitive to the threshold setting and yields state-of-the-art performance in accurate mangrove forest mapping.**

***Index Terms*— Convex optimization (CO), hyperspectral image, mangrove mapping (MM), Sentinel-2 image, small-data learning.**

I. Introduction

ACCURATE mangrove mapping (MM) in the tropical and subtropical regions is of paramount importance in the satellite remote sensing area because mangrove forests have a large capacity of above-ground carbon storage among the blue carbon ecosystems. Developing related technologies is, hence, critical for mitigating the climate change effects as pursued under the program of Reducing Emission from Deforestation and Degradation (REDD) [1]. In fact, among the key blue carbon ecosystems (i.e., coastal systems containing carbon storage, typically composed of mangroves, seagrasses, and saltmarshes), the mangrove forests have the greatest above-ground biomass and carbon storage, and have provided habitats for various species [2], [3], [4]. However, over the past century, the areas of mangrove forests have been decreasing, mainly owing to human activities and natural disasters, making the study of automatic change detection of the mangrove areas critically important. Therefore, designing a fast mangrove pixel classification algorithm, hopefully under the convex analysis framework for better algorithmic stability, becomes an essential research topic for effectively monitoring the structure and dynamics of mangrove forests via satellite remote sensing.

Remote sensing technology is necessary for efficiently obtaining mangrove distribution data because mangroves are often located in intertidal zones, making direct data acquisition almost infeasible. Optical satellite data are particularly useful for this target application due to the longer temporal coverage and good data accessibility. To effectively achieve the MM mission, hyperspectral remote sensing is definitely a critical technique due to the strong material identifiability of hyperspectral data [5], [6]. To this end, remotely sensed images acquired by hyperspectral satellites contain informative spectral signature shapes for facilitating the subsequent classification task. For example, the National Aeronautics and Space Administration (NASA) Earth Observing-1 (EO-1) satellite [7] covers a wide range of spectral wavelengths from visible-to-shortwave infrared (VSWIR) regions. However, as far as we have observed from existing remote sensing literature, the MM mission is mostly done by multispectral satellites. Sentinel-2 satellite, launched by the European Space Agency (ESA), is suitable for the MM task as it involves those characteristic spectral bands for depicting mangrove multispectral signatures, including the green band, the near-infrared (NIR) band, and the shortwave infrared (SWIR) band [8], [9]. Another advantage of adopting the Sentinel-2 data is that there are many well-developed off-the-shelf Sentinel-2 super-resolution software [10] for upsampling the coarser bands to achieve the highest spatial resolution, allowing us to conveniently

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TABLE I

SUMMARY OF DIFFERENT VEGETATION INDICES, WHERE THE NOTATIONS IN THE FOURTH COLUMN (MATHEMATICAL DEFINITION) ARE BASED ON THOSE DEFINED IN SECTIONS III-C AND III-D WITH i BEING THE PIXEL INDEX [9], [12], [13], [14]; THE ASSOCIATIONS BETWEEN THE WAVELENGTHS (IN THE THIRD COLUMN) AND THEIR CORRESPONDING BAND INDICES (IN THE FOURTH COLUMN) ARE AVAILABLE IN TABLE II

Index	Abbreviation	Formula	Mathematical Definition
Normalized Difference Vegetation Index	NDVI	(NIR-Red)/(NIR+Red)	$\frac{Z_{8,i}-Z_{4,i}}{Z_{8,i}+Z_{4,i}}$
Normalized Difference Water Index	NDWI	(Green-NIR)/(Green+NIR)	$\frac{Z_{3,i}-Z_{8,i}}{Z_{3,i}+Z_{8,i}}$
Modified Normalized Difference Water Index	MNDWI	(Green-SWIR1)/(Green+SWIR1)	$\frac{Z_{3,i}-Z_{11,i}}{Z_{3,i}+Z_{11,i}}$
Soil-Adjusted Vegetation Index	SAVI	$(1+L)$(NIR-Red)/(NIR+Red+L)	$\frac{(1+L)(Z_{8,i}-Z_{4,i})}{Z_{8,i}+Z_{4,i}+L}$
Modular Mangrove Recognition Index*	MMRI	(|MNDWI|-|NDVI|)/(|MNDWI|+|NDVI|)	$\frac{\lvert(Z_{3,i}-Z_{11,i})/(Z_{3,i}+Z_{11,i})\rvert-\lvert(Z_{8,i}-Z_{4,i})/(Z_{8,i}+Z_{4,i})\rvert}{\lvert(Z_{3,i}-Z_{11,i})/(Z_{3,i}+Z_{11,i})\rvert+\lvert(Z_{8,i}-Z_{4,i})/(Z_{8,i}+Z_{4,i})\rvert}$
Mangrove Vegetation Index*	MVI	(NIR-Green)/(SWIR1-Green)	$\frac{Z_{8,i}-Z_{3,i}}{Z_{11,i}-Z_{3,i}}$

* denotes mangrove-specific spectral indices.

use the Sentinel-2 data of uniform resolution to depict the mangrove's multispectral signatures. We remark that such a super-resolution task is nontrivial because most coarser bands are disjointed from those 10-m bands in their spectral ranges. On the other hand, it appears that most benchmark mangrove classification methods perform MM directly without super-resolving the Sentinel-2 image [9], preventing them from fully exerting the potential of Sentinel-2 MM. Except for hyperspectral and multispectral imagery, recent literature has explored the benefits and limitations of various high-resolution optical imaging techniques for MM, including synthetic aperture radar (SAR), light detection and ranging (LiDAR), and aerial imagery [11]. In Section II, various MM methodologies will be introduced sequentially.

II. RELATED WORK

A. Index-Based Methods

We observed that benchmark MM methods are often quite sensitive to the threshold setting required in the classification stage, making their practical applicability limited. Vegetation mapping using spectral characteristics is generally referred to as vegetation index (VI), such as the normalized difference VI (NDVI) and soil-adjusted VI (SAVI) [15], [16]. The NDVI, defined in Table I, is commonly used to evaluate various vegetation objects [12]. The healthy green vegetation has the properties of low red-light reflectance and high NIR reflectance, hence corresponding to high NDVI values. The NDVI values corresponding to the water and built-up areas tend to be negative or nearly zero. A similar formula is called normalized difference water index (NDWI) [13], [17], whose definition can be found in Table I. The water body has strong absorbability and low radiation in the spectral ranges from visible to infrared wavelengths. NDWI is proposed to highlight open water features, eliminating the soil and terrestrial vegetation features. It adopts NIR and green light reflectance to enhance the presence of the water feature and is sensitive to the liquid water content in vegetation canopies. Considering that both NDVI and NDWI do not entirely mitigate the effect caused by the background soil reflectance, SAVI was proposed to correct NDVI for suppressing the influence of soil brightness by incorporating a soil-adjustment factor L [16]. The definition of SAVI can be found in Table I, where the value of L depends on the amount of the covered green vegetation. The transformation made in SAVI is based on shifting the origin in the virtual NIR-Red wavelength space to account for first-order soil background variations.

Because the aforementioned indices are not customized for the MM task, some researchers have further proposed other VI specifically for MM, as introduced later. For example, the mangrove recognition index (MRI) [18] is designed based on the greenness of vegetation and wetness information for both low- and high-tide levels, but collecting related information would not be an easy task. The mangrove forest index (MFI) [19] was also proposed to detect mangrove forest and coastal vegetation using Sentinel-2 data, where the red band and the shortwave infrared 2 (SWIR2) band were adopted to form the baseline. Pixels with average values of red edge bands above the baseline are then categorized into the mangrove forest region to enhance the difference between the mangrove and water background. Next, a recently proposed index, termed modular MRI (MMRI) [14], is also based on NDVI and further relies on the modified NDWI (MNDWI) [13] defined in Table I. The modification made in MNDWI is to replace the NIR band of NDWI with some middle infrared (MIR) bands, such as Landsat TM band 5 or Sentinel-2 SWIR1 band [13]. Finally, the most well-known method would be the mangrove VI (MVI) [9] that jointly uses Sentinel-2 green, NIR, and SWIR bands, as detailed in Table I, for accurately mapping the mangrove extent. MVI is designed to capture the difference in greenness between mangrove forests and terrestrial vegetation, and to unsupervisedly discriminate the distinct moisture of mangroves. A higher MVI value indicated a higher probability of a pixel being classified into the mangrove areas. However, all the abovementioned index-based methods have two common disadvantages, including the high sensitivity to the cutoff value setting, as well as the independent pixelwise detection ignoring the spatial continuity of mangrove distribution; both disadvantages will be ruled out in our method, as will be detailed in Section III.

B. Deep Learning-Based Methods

Surprisingly, we found that benchmark MM methods are all index-based ones, completely ignoring the spatially neighboring information on the one hand and quite sensitive to

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the threshold setting on the other hand. Deep learning is a straightforward solution to incorporate desired spatial information. The method proposed in [20] detects mangrove pixels using the Google Earth Engine (GEE) based on random forest and classification and regression trees (CART). More sophisticatedly, a fully convolutional network (FCN) [21] is widely used for pixel-based classification and is, thus, suitable for the mangrove classification mission. FCN-based deep learning methods include UNet [22] and segmentation network (SegNet) [23], which are, respectively, introduced in the following. UNet consists of a symmetric encoder–decoder architecture with skip connections. The encoder extracts features through a series of convolutional layers and downsampling layers, while the decoder upsamples the extracted features to the original input resolution. The skip connections connect each downsampling layer and the upsampling layer to fuse the high-resolution features from the encoder for promoting good spatial information recovery [22]. SegNet consists of an encoder and a decoder, followed by a pixelwise classification layer. The architecture of the encoder is identical to the 13 convolutional layers in VGG16 [24], whereas the decoder transforms the feature maps to input resolution for pixelwise classification. The decoder uses pooling indices, computed in the corresponding encoder max-pooling operation, to perform nonlinear upsampling [23]. It is noteworthy to mention other deep learning-based methods for MM, such as GC-UNet [25] and DCNN [26]. The GC-UNet approach exploits global context blocks (GCBs) to extract long-range dependency features under the multiscaled UNet framework [22] and then fuses the low-level features with high-level features through adaptively spatial feature fusion (ASFF) modules; in addition, the computational efficiency is improved by split-based convolutional operations (SPConvs). Besides, the DCNN method, as a commonly used basic module, comprises one convolution layer and two dense layers that can capture high-level features, followed by a fully connected layer for automatically detecting mangrove areas of Sentinel-2 and WorldView-2 images. From another perspective, Guo et al. [27] involved using various vegetation and water indices as input data cubes to train the ResNet-based mangrove extraction network (ME-Net). By substituting max-pooling layers with convolutional strides and dynamic routing [28], Guo et al. [29] proposed Capsules-UNet based on the basic UNet structure to generate more precise MM results.

C. Convex Optimization-Based Methods

Despite the widespread adoption of deep learning (DE) techniques, it is worth mentioning that there is literature that uses convex formulations to address challenges related to multiband classification or change detection tasks, especially under the scarcity of MM cases. Zheng et al. [30] examined the characteristics of change regions and subsequently redefined the problem of change detection using convex relaxation techniques, which was efficiently solved using the alternating direction method of multipliers (ADMM). Moreover, Liu et al. [31] employed convex vectorial total variation (TV) regularization (REG) to facilitate the piecewise smoothness of boundaries in the classification map for better performance.

In our proposed method, we will introduce CO into deep learning to conduct effective deep learning without relying on big data or math-heavy REG, as will be detailed in Section III. First, we concisely revisit the idea of the deep plug-and-play (PnP) approach adopted in various applications, as alluded in [32], [33], [34], [35], and [36]. In (1), the REG function REG(·) typically results in a proximal operator that acts as a denoiser. However, this operator can often be intricate and nonconvex, making it infeasible to define the operator explicitly or to solve it with closed-form solutions in ADMM. As an alternative, a DE-based denoiser is integrated into the ADMM to *approximately* implement the operator. Therefore, there is no theoretical assurance of convergence for the PnP algorithms since the implicit form of the regularizer makes the proximal operator not *exactly* solvable. Besides, a DE-based denoiser must exhibit superior efficacy to attain a precise solution, indicating the necessity of substantial data volume required during the training phase.

Considering that both big data collection and labeling in DE are difficult and time-consuming, this work further introduces convex optimization (CO) to solve the dilemma. More specifically, this article will adopt the CODE small-data learning/imaging theory [37], where the ADMM optimizer [38] in CO and the adaptive moment estimation (Adam) optimizer [39] in DE are blended to achieve high-dimensional satellite remote sensing tasks using ADMM-Adam optimization. For the first time, the CODE theory is adopted for the MM mission. Blending the advantages of CO and DE, the CODE theory solves the inverse problem of satellite image mapping/classification by inferring a rough mangrove signature from the optical satellite data (e.g., Sentinel-2 data and hyperspectral data) for deep REG. As stated in the recent literature [37], when employing deep REG under the CODE framework, the deep learning stage just needs to return a roughly estimated classification map. Though the rough small data solution is expected to be ineffective, it still contains critical information for the final learning criterion to work well. Such information is extracted using a simple convex function, known as $\boldsymbol{Q}$-quadratic norm [37], thereby allowing us to bypass math-heavy optimization procedures that would otherwise slow down the algorithmic speed. Considering that those mangrove forest distributions are often spatially continuous, a target pixel and the above rough spectral signature are spatially augmented to 3-D spatial–spectral tensors for exploiting such a spatial continuity, followed by feeding the tensors into a customized Siamese deep network [40] for roughly judging whether the target pixel belongs to the mangrove area or not. We then design a learning criterion for the optical satellite classification mission by incorporating the above information into a convex REG scheme and apply the so-called convex relaxation technique to the problem domain, transforming the learning criterion into the convex analysis framework for efficient computation. As demonstrated from extensive experiments, our convex deep MM (CODE-MM) algorithm achieves state-of-the-art (SOTA) classification performance in rapid and accurate MM.

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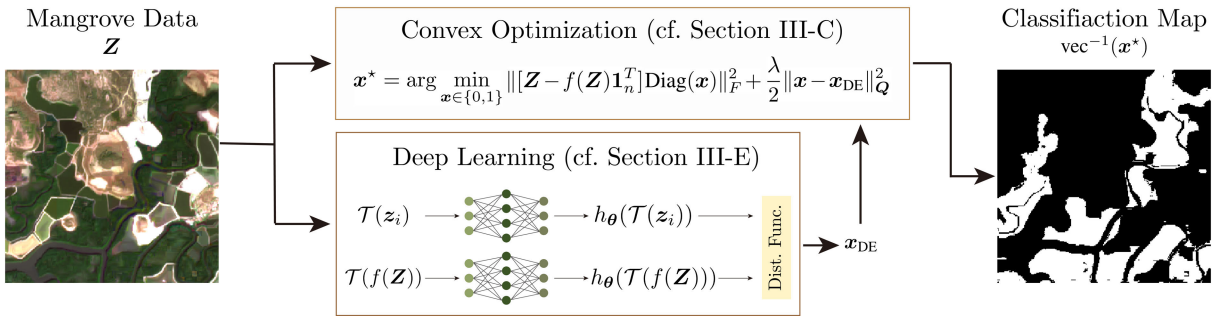


Fig. 1. Overview of the proposed CODE-MM, showing how the deep solution $\boldsymbol{x}_{\mathrm{DE}}$ is incorporated into the convex regularizer.

The remainder of this article is organized as follows. The detailed description of the proposed method is introduced in Section III. This part involves a brief recall of the CODE theory [37], the formulation of the considered Sentinel-2 classification problem, and the details for both the convex optimization and deep learning methods. The formulation is also easily generalized to the case of hyperspectral satellites. Experimental results will be summarized and discussed in Section IV. Finally, conclusions are drawn in Section V.

Some standard notations are collectively defined here for better readability. $\mathbb{R}^n$ is the n-dimensional Euclidean space, $\mathbb{R}^{m\times n}$ is the $(m\times n)$-dimensional real-valued matrix space, and $\mathbb{R}^{m\times n\times l}$ is the $(m\times n\times l)$-dimensional real-valued three-way tensor space. $\mathbf{1}_N$ denotes the all-one N-vectors. The notation $\mathrm{conv}(\cdot)$ denotes a convex hull. $\|\cdot\|_F$ and $\|\cdot\|_2$ are the Frobenius norm and ℓ_2-norm, respectively. $\mathrm{vec}_{m\times n}^{-1}(\boldsymbol{v})$ is the $m\times n$ matrix $\boldsymbol{M}$ satisfying $\mathrm{vec}(\boldsymbol{M}) = \boldsymbol{v}$, where $\mathrm{vec}(\boldsymbol{M})$ is the vector formed by sequentially stacking the columns in $\boldsymbol{M}$. $\mathrm{Diag}(\boldsymbol{v})$ is a diagonal matrix whose ith diagonal entry is $[\boldsymbol{v}]_i$, i.e., the ith entry of $\boldsymbol{v}$, for a given vector $\boldsymbol{v}$.

III. Optical Satellite Mangrove Mapping via Convex Optimization and Deep Learning

The proposed algorithm, termed convex deep mangrove mapping (CODE-MM), is mainly developed for the Sentinel-2 satellite, the mainstream satellite in MM, as it involves those key green/infrared bands for characterizing mangrove multispectral signatures. We will also generalize CODE-MM to the hyperspectral case, which should be the trend for various classification-related satellite missions in the future due to the strong material identifiability of hyperspectral images.

In Section III-A, we first recall the CODE theory [37] explaining how it helps software developers to avoid math-heavy optimization and time-consuming big data collection. In Section III-B, we discuss a super-resolution technique that enables the classification of Sentinel-2 data at the highest resolution. In Section III-C, we formally formulate the Sentinel-2 image classification problem under the convex analysis framework with the target application to MM. In Section III-D, we propose an automatic computing method to roughly estimate the mangrove multispectral signature for subsequent deep REG. The corresponding implementation details, as well as the extension to the hyperspectral case, will then be presented in Section III-E. In Sections III-A–III-E, we will develop the CODE-MM technique, as graphically illustrated in Fig. 1.

A. CODE Small-Data Learning Theory

Classification is a challenging inverse problem, for which the unknown quantity and the observation are the classification labeling and the optical satellite data, respectively. As for how to establish the corresponding data-fitting (DF) and REG terms for this inverse problem, it will be greatly detailed in Sections III-C and III-E. The two most popular theories for solving inverse problems would be CO and deep learning (DE). However, CO usually induces heavy math, while DE relies on the existence of big data; note that big data are often lacking in remote sensing as high-quality satellite images are expensive and hard to obtain. In recent remote sensing literature [37], a new theoretical framework has been proposed to blend the advantages of the ADMM optimizer in CO [41] and the Adam optimizer in DE [39]. This novel framework, termed CODE, has several advantages:

1) It avoids mathematically sophisticated REG so that the software designers can avoid math-heavy optimization procedures.
2) It works well even just with small data, quite appealing in satellite remote sensing; occasionally, it may still work well when only single data are available [37].
3) It does not rely on an advanced DE network structure, so it is friendly to general researchers (rather than manageable only to experienced DE researchers). Empirically, a naive DE network structure is sufficient, as we have observed that a rough DE solution is sufficient (under the CODE framework) for effective REG.

The CODE theory [37] is briefly recalled in the following to illustrate how it avoids math-heavy REG even in the absence of big data.

CODE proposed to solve imaging inverse problems by adopting the following *convex* learning criterion:

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}\in\mathcal{S}} \ \mathrm{DF}(\boldsymbol{x}) + \lambda \mathrm{REG}(\boldsymbol{x}), \tag{1}$$

where the data-fitting term $\mathrm{DF}(\boldsymbol{x})$ is to yield a valid solution $\boldsymbol{x}^\star$ (i.e., the target image echoing the observed data), the regularizer $\mathrm{REG}(\boldsymbol{x})$ is to force some desired physical properties to the solution $\boldsymbol{x}^\star$, $\mathcal{S}$ is the problem domain, and $\lambda > 0$ controls the REG strength. Note that $\boldsymbol{x}$ is an optimization variable to be computed to well balance the DF and REG terms. In CODE [37], the regularizer is designed to be a very simple differentiable convex function given by

$$\mathrm{REG}(\boldsymbol{x}) \triangleq \frac{1}{2}\|\boldsymbol{x} - \boldsymbol{x}_{\mathrm{DE}}\|_{\boldsymbol{Q}}^2, \tag{2}$$

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Algorithm 1 ADMM Algorithm

Given $c > 0$ and $\boldsymbol{x}_{\mathrm{DE}}$. Initialize $q := 0$, $\boldsymbol{d}^q$, and $\boldsymbol{x}^q$.
repeat
  update $\boldsymbol{z}^{q+1} := \arg\min_{z\in\mathcal{S}} \mathcal{L}(\boldsymbol{x}^q, \boldsymbol{z}, \boldsymbol{d}^q)$;
  update $\boldsymbol{x}^{q+1} := \arg\min_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}, \boldsymbol{z}^{q+1}, \boldsymbol{d}^q)$;
  update $\boldsymbol{d}^{q+1} := \boldsymbol{d}^q - \left(\boldsymbol{x}^{q+1} - \boldsymbol{z}^{q+1}\right)$;
  $q := q + 1$;
until the predefined stopping criterion is met.
Output $\boldsymbol{x}^\star := \boldsymbol{x}^q$.

where the $\boldsymbol{Q}$-quadratic norm of a given vector $\boldsymbol{v}$ (with $\boldsymbol{Q}$ being a positive semidefinite (PSD) matrix) is defined as $\|\boldsymbol{v}\|_{\boldsymbol{Q}}^2 \triangleq \boldsymbol{v}^T\boldsymbol{Q}\boldsymbol{v}$ [42]. Here, $\boldsymbol{x}_{\mathrm{DE}}$ can be regarded as a rough guess of the final solution $\boldsymbol{x}^\star$ and can be obtained from the Adam optimizer [39] based on some customized DE network structure. As indicated in [37], the proposed regularizer is actually more general than the prestigious Tikhonov regularizer [43]. Hence, it is not surprising that (2) is able to retain sufficient structural information for regularizing the final solution. For example, the frequently used self-similarity (SS) regularizer [10] is to encourage similar patches at different locations of the solution. Even when $\boldsymbol{x}_{\mathrm{DE}}$ is not good enough (e.g., somewhat blurred) due to the absence of big data, those similar patches in the small-data solution should still be similar in the (probably blurred) $\boldsymbol{x}_{\mathrm{DE}}$, and hence, by minimizing (2), the SS structure in $\boldsymbol{x}_{\mathrm{DE}}$ also propagates to the final solution. This explains why the regularizer (2) is sufficient to describe some desired structural properties. Another advantage must be the fast algorithmic computation brought by the simple mathematical expression of (2). As aforementioned, the CODE framework does not rely on big data or advanced DE network structure since we do not expect the DE network to return an accurate solution. Instead, a roughly guessed solution $\boldsymbol{x}_{\mathrm{DE}}$ containing some useful property/structure is what we need for the subsequent regularizer design, so we do not expect $\boldsymbol{x}_{\mathrm{DE}}$ to be a good solution to the target inverse problem. However, the rough solution $\boldsymbol{x}_{\mathrm{DE}}$ still contains useful information learned from small data [37]. Therefore, the $\boldsymbol{Q}$-quadratic norm $\|\cdot\|_{\boldsymbol{Q}}$ in (2) can be viewed as a simultaneous feature extraction and fusion operator (SFEFO), where the SFEFO $\|\cdot\|_{\boldsymbol{Q}}$ extracts useful information embedded in $\boldsymbol{x}_{\mathrm{DE}}$ and fuses it into the final solution $\boldsymbol{x}^\star$ by minimizing (2) [37].

We remark that, unlike the non-differentiable TV regularizer or the sophisticated graph-embedded SS regularizer [10], the proposed regularizer (2) has a very simple math form, thereby leading to a very simple and fast algorithm for implementing criterion (1). Although TV and SS regularizers are also convex functions [10], their mathematical forms are more complicated, inducing a math-heavy optimization procedure that is not friendly/manageable to general researchers and software engineers. In [37], the matrix $\boldsymbol{Q}$ is simply designed using a basic feature extractor known as principal component analysis (PCA).

Finally, under the CODE framework, the criterion (1) is implemented using the ADMM optimizer [41] that is computationally very efficient for handling large-scale convex problem, as detailed in Algorithm 1, where the augmented Lagrangian is defined as

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{d}) \triangleq \mathrm{DF}(\boldsymbol{x}) + \lambda \mathrm{REG}(\boldsymbol{z}) + \frac{c}{2}\|\boldsymbol{x} - \boldsymbol{z} - \boldsymbol{d}\|_2^2, \quad (3)$$

in which the penalty parameter $c := 0.001$, $\boldsymbol{x}$ and $\boldsymbol{z}$ are the primal variables (i.e., variables in the original space), and $\boldsymbol{d}$ is the dual variable (i.e., the variable in the Lagrange dual space). One may refer to [37, Sec. II-A] for a concise introduction to ADMM, where one would realize that the use of ADMM is simple and practical; simply speaking, the dual decomposability of ADMM allows for solving a rather complicated optimization problem via solving a series of relatively simple subproblems, as observed in several ADMM-based imaging inverse algorithms [44], [45]. As a quick summary, CODE exploits Adam optimizer to obtain a convex regularizer of very simple math form via small-data DE and then solves the regularized convex learning criterion using ADMM optimizer.[1] In this work, without million-scale big data, the CODE framework will be applied to solve the classification problem for the first time, and as it turns out, it achieves remarkable classification performance even just with hundred-scale small data.

TABLE II
DETAILED SPECTRAL INFORMATION OF SENTINEL-2 BANDS [48]

Band	Wavelength range (nm)	Resolution (m)
B1 Coastal Aerosol	443-453	60
B2 Blue (B)	458-523	10
B3 Green (G)	543-578	10
B4 Red (R)	650-680	10
B5 Red Edge 1 (RE1)	698-713	20
B6 Red Edge 2 (RE2)	733-748	20
B7 Red Edge 3 (RE3)	773-793	20
B8 Near Infrared (NIR)	785-900	10
B8a Near Infrared Narrow (NIRn)	855-875	20
B9 Water Vapour	935-955	60
B10 Shortwave Infrared/Cirrus	1360-1390	60
B11 Shortwave Infrared 1 (SWIR1)	1565-1655	20
B12 Shortwave Infrared 2 (SWIR2)	2100-2280	20

B. Spatial Super-Resolution by Sentinel-2 Super-Resolution via Scene-Adapted Self-Similarity (SSSS)

In this section, we formulate the Sentinel-2 mangrove classification problem based on the CODE theory. As can be seen from Table II, the formulation of the Sentinel-2 classification problem is tricky as different spectral bands have different spatial resolutions. Specifically, after excluding band 10 (containing cirrus information just for atmospheric correction), the Sentinel-2 satellite involves $\ell = 12$ spectral bands, including $\ell_1 = 4$ high-resolution bands with 10-m ground sampling distances (GSD[2]), $\ell_2 = 6$ medium-resolution bands with 20 m GSD, and $\ell_6 = 2$ low-resolution bands with 60-m GSD [47].

[1] One may wonder how such a simple CODE theory can effectively handle highly challenging inverse problems like hyperspectral image inpainting. We refer interested readers to [37, Sec. II-B] for a comprehensive explanation from both theoretical and experimental perspectives.

[2] We refer interested readers to [46] for the definition of GSD.

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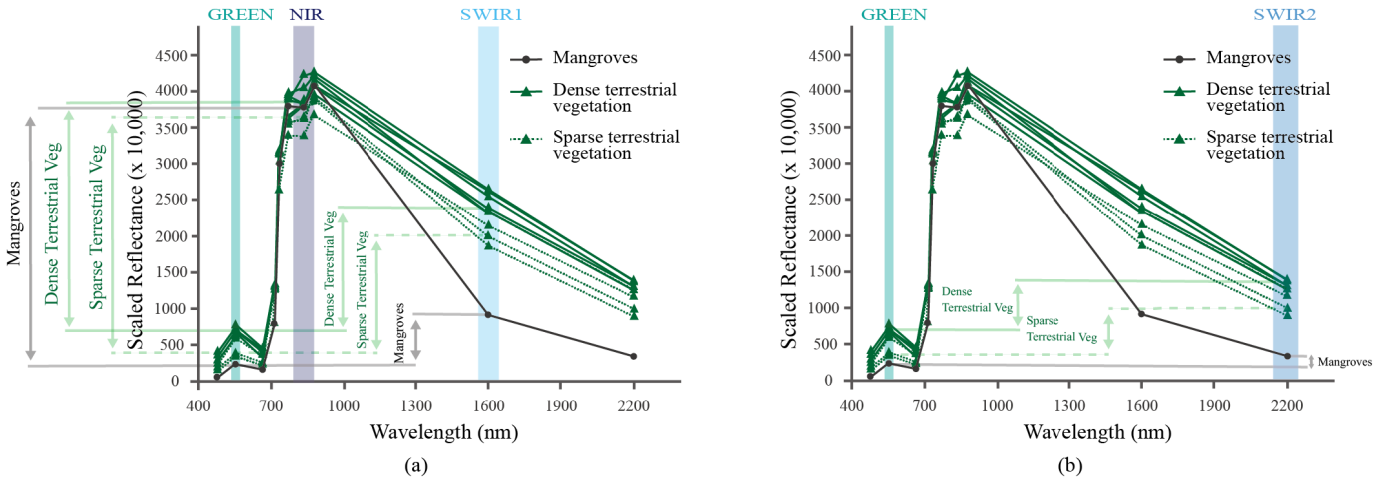


Fig. 2. Specific characteristics of the mangrove's Sentinel-2 multispectral signature mainly consist in the green, NIR, and SWIR bands, where NIR and SWIR bands characterize the vegetation greenness and the water absorption (moisture), respectively. To see the characteristics of the mangrove's signature, the difference of reflectance value between the NIR band B8 (resp., SWIR1 band B11) and the green band B3 is larger (resp., smaller) than that observed in other terrestrial vegetation signatures, as illustrated in (a), where the multispectral signatures are replicated from the remote sensing literature [9] for self-contained purpose. To enhance the capability of discriminating the distinct moisture of mangrove forests, the Sentinel-2 SWIR2 band is further introduced because the difference of reflectance values between the SWIR2 band B12 and the green band B3 is also relatively smaller than that observed in typical terrestrial vegetation signatures, as illustrated in (b).

We aim to achieve the classification of the highest resolution (i.e., 10-m), so we first super-resolve all the $\ell = 12$ bands to have a 10-m spatial resolution. This super-resolution task can be achieved by using the SOTA algorithm, known as Sentinel-2 super-resolution via SSSS [10]. The SSSS algorithm defines the widely known SS regularizer (early rooted in the computer vision area [49]) as a convex function and efficiently solves the corresponding convex problem by well exploiting a smart graph learning and the so-called block-circulant-circulant-block matrix structure [10]. Now, as all the spectral bands have the same number of pixels, we can conveniently represent the super-resolved Sentinel-2 image using a matrix $\boldsymbol{Z} = [\boldsymbol{z}_1, \ldots, \boldsymbol{z}_n] \in \mathbb{R}^{\ell \times n}$ for elegant formulation, where $n = n_1 n_2$ is the number of pixels in the $n_1 \times n_2$ super-resolved image, and the $\ell = 12$ rows in $\boldsymbol{Z}$ correspond to the $\ell = 12$ super-resolved band images.

C. Formulation of the Sentinel-2 Classification Problem

To customize the DF term in (1) for the Sentinel-2 mangrove classification problem, we define the vector

$$\boldsymbol{x} = [x_1, \ldots, x_n]^T \in \mathcal{D} \triangleq \{0, 1\}^n \tag{4}$$

where $x_i = 1$ if the ith pixel $\boldsymbol{z}_i$ is classified into the mangrove area (otherwise, $x_i = 0$). We also assume that the multispectral signature vector of mangrove pixels is available or computable from the super-resolved Sentinel-2 image $\boldsymbol{Z}$, so we denote the signature vector as a function of $\boldsymbol{Z}$, i.e., $f(\boldsymbol{Z}) \in \mathbb{R}^\ell$. A technique for automatically computing the signature $f(\boldsymbol{Z}) \in \mathbb{R}^\ell$ will be detailed in Section III-D. Therefore, we employ the CODE theory to design the Sentinel-2 mangrove classification criterion in the form of (1) as follows. As the REG form $\mathrm{REG}(\boldsymbol{x})$ has been presented in (2), we just focus on specifying the DF term $\mathrm{DF}(\boldsymbol{x})$ in (1), i.e.,

$$\mathrm{DF}(\boldsymbol{x}) \triangleq \|[\boldsymbol{Z} - f(\boldsymbol{Z})\mathbf{1}_n^T]\,\mathrm{Diag}(\boldsymbol{x})\|_F^2, \tag{5}$$

whose physical meaning is illustrated in the following. By penalizing the DF term (5), only those x_i satisfying $\boldsymbol{z}_i \approx f(\boldsymbol{Z})$ have a good chance to be classified as mangrove pixels (i.e., $x_i = 1$) in order to prevent the criterion (1) returning a large objective value. In other words, minimization of the DF term (5) can avoid a non-mangrove pixel $\boldsymbol{z}_i$ to be misclassified as a mangrove pixel.

To complete the design of the Sentinel-2 mangrove classification criterion (1), the last task is to mention the problem domain, which is naturally chosen as $\mathcal{S} \triangleq \mathrm{conv}\,\mathcal{D}$, which is the so-called convex relaxation trick to cast the criterion into the CO framework [30]. The physical meaning of $\mathcal{S} \triangleq \mathrm{conv}\,\mathcal{D}$ is to generalize the binary classification map $\boldsymbol{x}$ (with values in $\{0, 1\}$) to a probability map $\boldsymbol{x}$ (with values in $[0, 1]$), where the latter returns a value x_i that can be regarded as the chance of $\boldsymbol{z}_i$ being a mangrove pixel; in case a binary map is preferred, one can simply set a threshold [30]. Based on the DF term and regularizer presented above, the detailed implementation of our Sentinel-2 mangrove classification criterion (1) (namely, deriving closed-form solutions for Steps 3 and 4 of Algorithm 1), together with the implementation of the deep regularizer, will be collectively presented in Section III-E.

D. Automatic Computing of Mangrove Multispectral Signature

In this section, we develop the function f mentioned in Section III-C for automatically computing the multispectral signature $f(\boldsymbol{Z})$ of mangrove pixels from the super-resolved Sentinel-2 image $\boldsymbol{Z}$. Since computing $f(\boldsymbol{Z})$ is just an intermediate step of our classification algorithm, we aim to develop a fast computing method. To this end, we propose to roughly identify those mangrove-like pixels from the super-resolved Sentinel-2 image $\boldsymbol{Z}$, followed by averaging these pixels to estimate the mangrove multispectral signature

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$f(\boldsymbol{Z})$; this approach avoids the difficulty of seeking a good universal threshold value required in other index-based methods, as will be further illustrated in our experimental investigation [cf. (7)].

Employing the specific characteristics of mangrove's Sentinel-2 multispectral signature, the first observation is that the difference between the NIR reflectance value (cf. band 8 in Table II) and the green reflectance value (cf. band 3 in Table II) is always larger than the difference observed in other terrestrial vegetation signatures, as reported in the recent remote sensing literature [9]; cf. Fig. 2(a) for a graphical illustration. On the other hand, the second observation for mangrove's multispectral signature is that the difference of reflectance values between the SWIR1 band (cf. band 11 in Table II) and the green band (cf. band 3 in Table II) is relatively smaller than the difference observed in either dense or sparse terrestrial vegetation signatures [9]; cf. Fig. 2(a) for a graphical illustration. As for the physical meaning, the first observation is to capture the difference in greenness between mangrove forest and terrestrial vegetation, while the second observation is to unsupervisedly discriminate the distinct moisture of mangroves [9]. Accordingly, a natural way to identify a mangrove pixel $\boldsymbol{z}_i$ is to see whether the corresponding index value of $(\boldsymbol{Z}_{8,i} - \boldsymbol{Z}_{3,i})/(\boldsymbol{Z}_{11,i} - \boldsymbol{Z}_{3,i})$ has been greater than a suitable threshold or not. As a result, a larger index value indicates a higher chance of $\boldsymbol{z}_i$ being a mangrove pixel, and a default setting of the threshold value is 4.5, as suggested in [9].

Though the above judicious observations are quite helpful for us to characterize the signature $f(\boldsymbol{Z})$, setting a good threshold value is very difficult and could be data-dependent. To address the difficulty, we propose a method that is less sensitive to the threshold setting. Specifically, under the premise mentioned at the beginning of Section III-D, we just need to average those roughly identified mangrove-like pixels from $\boldsymbol{Z}$, allowing us to identify just part of those mangrove-like pixels (rather than all of them). Thus, we can set a safer (namely, higher) threshold value; in this work, it is fixed as high as the data-independent value of $\alpha := 6$ in order to strictly collect high-quality mangrove-like pixels for the subsequent high-precision averaging step. Moreover, to enhance the capability of discriminating the distinct moisture of mangroves, we further introduce the Sentinel-2 SWIR2 band (cf. band 12 in Table II) to upgrade the index; note that leaf liquid water affects the reflectance values in the SWIR bands. To be specific, we observe that, for the mangrove's multispectral signature, the difference of reflectance values between the SWIR2 band and the green band is also relatively smaller than that observed in typical terrestrial vegetation signature; cf. Fig. 2(b) for a graphical illustration. This observation helps more safely identify those mangrove-like pixels from $\boldsymbol{Z}$ after incorporating such a difference into the denominator of the index.

To summarize, we define an SWIR-enhanced MVI (SEMVI) as

$$g(\boldsymbol{z}_i) \triangleq \frac{\boldsymbol{Z}_{8,i} - \boldsymbol{Z}_{3,i}}{(\boldsymbol{Z}_{11,i} - \boldsymbol{Z}_{3,i}) + (\boldsymbol{Z}_{12,i} - \boldsymbol{Z}_{3,i})}. \tag{6}$$

Therefore, we have achieved the automatic computing of the mangrove's multispectral signature as

$$f(\boldsymbol{Z}) \triangleq \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \boldsymbol{z}_i, \tag{7}$$

where $\mathcal{M} \triangleq \{i \mid g(\boldsymbol{z}_i) \geq \alpha,\ i = 1, \ldots, n\}$ is the set of those identified mangrove-like pixels in $\boldsymbol{Z}$, and here, $|\mathcal{M}|$ denotes the set cardinality of $\mathcal{M}$. We will experimentally show that the setting of the threshold α has less impact on the classification performance, and the automatically computed $f(\boldsymbol{Z})$ does yield a high-efficacy classification criterion (1) under the CODE framework.

E. Implementation Details, Deep Regularizer Design, and Extension of CODE-MM to the Hyperspectral Case

We now present the implementation details of the proposed CODE-based Sentinel-2 mangrove classification criterion (1), including the computation for Steps 3 and 4 of Algorithm 1 and the computation of the rough solution $\boldsymbol{x}_{\mathrm{DE}}$ required in the deep regularizer (2). At the end of this section, we then extend CODE-MM to the hyperspectral case.

As we have recalled the great advantages of the CODE theory at the beginning of Section III-A, it does not involve a math-heavy optimization procedure, and a naive DE network structure is sufficient for CODE to work well even we just have small data, making the theory highly friendly to general researchers, as will be observed in the following [37].

We first derive the closed-form solution for the subproblem in Step 3 of Algorithm 1. Let Π be the projection operator onto the interval $[0, 1]$, i.e., $\Pi(y) = y$ for $y \in [0, 1]$, $\Pi(y) = 1$ for $y > 1$, and $\Pi(y) = 0$ for $y < 0$; if the input is a vector $\boldsymbol{y}$, then the projector Π applies to the vector $\boldsymbol{y}$ componentwisely. By combining (2), (3), and (5), and removing terms independent of $\boldsymbol{z}$, we have

$$\begin{aligned}
\boldsymbol{z}^{q+1} &:= \arg \min_{\boldsymbol{z} \in \mathrm{conv}\mathcal{D}} \mathcal{L}(\boldsymbol{x}^q, \boldsymbol{z}, \boldsymbol{d}^q) \\
&= \arg \min_{\boldsymbol{z} \in \mathrm{conv}\{0,1\}^n} \frac{\lambda}{2} \|\boldsymbol{z} - \boldsymbol{x}_{\mathrm{DE}}\|_{\boldsymbol{Q}}^2 + \frac{c}{2} \|\boldsymbol{x}^q - \boldsymbol{z} - \boldsymbol{d}^q\|_2^2 \\
&= \Pi\left[\frac{\lambda}{\lambda + c} \boldsymbol{x}_{\mathrm{DE}} + \frac{c}{\lambda + c} (\boldsymbol{x}^q - \boldsymbol{d}^q)\right],
\end{aligned}$$

where we have used the fact that the classification map involves just one channel, thereby implying that the 1-D PCA operator performs nothing but the identity mapping, i.e., $\|\boldsymbol{v}\|_{\boldsymbol{Q}}^2 = \boldsymbol{v}^T \boldsymbol{Q} \boldsymbol{v} = \boldsymbol{v}^T \boldsymbol{I} \boldsymbol{v} = \|\boldsymbol{v}\|_2^2$; note that this is true only for the single channel scenario like when studying the classification problem. Next, we derive the closed-form solution for the subproblem in Step 4 of Algorithm 1. Again, by combining (2), (3), and (5), and removing terms independent of $\boldsymbol{x}$, we then have

$$\begin{aligned}
\boldsymbol{x}^{q+1} &:= \arg \min_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}, \boldsymbol{z}^{q+1}, \boldsymbol{d}^q) \\
&= \arg \min_{\boldsymbol{x}} \|[\boldsymbol{Z} - f(\boldsymbol{Z}) \boldsymbol{1}_n^T] \, \mathrm{Diag}(\boldsymbol{x})\|_F^2 \\
&\quad + \frac{c}{2} \|\boldsymbol{x} - \boldsymbol{z}^{q+1} - \boldsymbol{d}^q\|_2^2,
\end{aligned}$$

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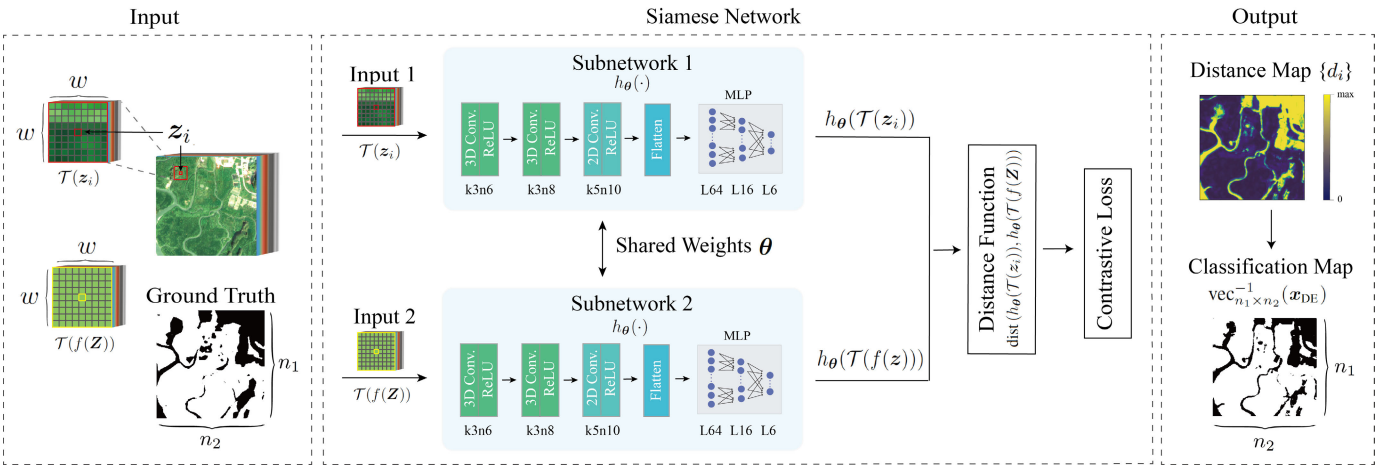


Fig. 3. Graphical illustration of the proposed Siamese network structure $h_{\boldsymbol{\theta}}(\cdot)$. The two hyperparameters (s, c) in "k(s)n(c)" mean that the corresponding layer involves c channels with a kernel size of s, while the hyperparameter r in "L(r)" means that the corresponding MLP layer involves r neurons. Unlike conventional Siamese networks, the network inputs are extended to 3-D tensors [i.e., $\mathcal{T}(\boldsymbol{z}_i)$ and $\mathcal{T}(f(\boldsymbol{Z}))$] for well including both spatial and spectral information in the learning procedure. Then, the distance between the two output feature vectors [i.e., $h_{\boldsymbol{\theta}}(\mathcal{T}(\boldsymbol{z}_i))$ and $h_{\boldsymbol{\theta}}(\mathcal{T}(f(\boldsymbol{Z})))$] is computed to judge whether the pixel $\boldsymbol{z}_i$ is similar to the mangrove signature $f(\boldsymbol{Z})$ or not.

from which one can easily verify that the solution to the unconstrained problem is

$$[\boldsymbol{x}^{q+1}]_i := \frac{c[\boldsymbol{z}^{q+1} + \boldsymbol{d}^q]_i}{2\|\boldsymbol{z}_i - f(\boldsymbol{Z})\|_2^2 + c}, \quad \forall i = 1, \ldots, n.$$

Therefore, we have completed the derivation of Algorithm 1.

Next, we present how the rough solution $\boldsymbol{x}_{\mathrm{DE}}$ required in the deep regularizer (2) is obtained. With the availability of the multispectral signature $f(\boldsymbol{Z})$ of mangrove pixels (cf. Section III-D), the Siamese network [50] is suitable for designing the deep regularizer in the CODE framework, as explained in the following. The Siamese network, originally introduced for signature verification [50], has found numerous applications, including face verification, image recognition, object tracking, and hyperspectral analysis [51], [52], [53], [54]. As its name suggests, the Siamese network involves two exactly the same subnetworks for distinguishing whether the two inputs are similar or not. Under this dual structure, the mangrove classification task involves the subnetworks to identify whether a pixel $\boldsymbol{z}_i$ is a mangrove pixel by computing the similarity score between $\boldsymbol{z}_i$ and the fixed mangrove signature $f(\boldsymbol{Z})$. To facilitate understanding of the proposed CODE-MM, we first provide a detailed account of: 1) the design of the network structure; 2) the training process of the network; and 3) the way we connect the network result $\boldsymbol{x}_{\mathrm{DE}}$ to the deep regularizer.

The adopted Siamese network structure is detailed in Fig. 3, where the notation "k(s)n(c)" means that the corresponding layer involves c channels with a 2-D kernel of size $s \times s$ or with a 3-D kernel of size $s \times s \times s$. Since the two subnetworks have the same structure in the Siamese network, let us denote the subnetwork as a function $h_{\boldsymbol{\theta}}(\cdot)$, where $\boldsymbol{\theta}$ is the parameters shared by both subnetworks. As aforementioned, our aim is to evaluate the similarity between $f(\boldsymbol{Z})$ and $\boldsymbol{z}_i$; both are 1-D vectors involving just spectral dimension. However, directly evaluating the similarity using 1-D inputs does not well exploit the spatial continuity of typical mangrove areas; to evaluate whether the pixel $\boldsymbol{z}_i$ is similar to the mangrove signature $f(\boldsymbol{Z})$ or not, it is better to further consider the information of pixels spatially adjacent to $\boldsymbol{z}_i$. Thus, our proposed Siamese network exploits the information embedded in the two spatial dimensions by augmenting the 1-D input vectors [i.e., $\boldsymbol{z}_i$ and $f(\boldsymbol{Z})$] to 3-D tensors, denoted as $\mathcal{T}(\boldsymbol{z}_i)$ and $\mathcal{T}(f(\boldsymbol{Z}))$. Here, the multispectral signature $\boldsymbol{z}_i \in \mathbb{R}^{12}$ is mapped by the operator "$\mathcal{T} : \mathbb{R}^{12} \to \mathbb{R}^{w \times w \times 12}$" to a 3-D cube $\mathcal{T}(\boldsymbol{z}_i) \in \mathbb{R}^{w \times w \times 12}$ formed by the continuous $w \times w$ pixels in the 12-band Sentinel-2 image $\boldsymbol{Z}$ with $w := 9$ being the window size, where $\boldsymbol{z}_i$ is the center pixel (i.e., the (5, 5)th pixel) of the 3-D cube $\mathcal{T}(\boldsymbol{z}_i) \in \mathbb{R}^{9 \times 9 \times 12}$; based on (7), the 3-D cube $\mathcal{T}(f(\boldsymbol{Z}))$ is then defined as

$$\mathcal{T}(f(\boldsymbol{Z})) = \mathcal{T}\left(\frac{1}{|\mathcal{M}|}\sum_{i \in \mathcal{M}} \boldsymbol{z}_i\right) = \frac{1}{|\mathcal{M}|}\sum_{i \in \mathcal{M}} \mathcal{T}(\boldsymbol{z}_i).$$

Each subnetwork $h_{\boldsymbol{\theta}}(\cdot)$ in the proposed Siamese network contains three convolution layers, one flatten layer, and one multilayer perceptron (MLP), as illustrated in Fig. 3. We remark that we add a 2-D convolution layer after the 3-D convolution because it has been reported that this trick enhances spatial feature learning with fewer computing resources [54]. Accordingly, the proposed Siamese network outputs two feature vectors, i.e., $h_{\boldsymbol{\theta}}(\mathcal{T}(\boldsymbol{z}_i))$ and $h_{\boldsymbol{\theta}}(\mathcal{T}(f(\boldsymbol{Z})))$, whose similarity will then be evaluated using the Euclidean distance $\mathrm{dist}(\boldsymbol{v}_1, \boldsymbol{v}_2) = \|\boldsymbol{v}_1 - \boldsymbol{v}_2\|_2$.

To train the Siamese network $h_{\boldsymbol{\theta}}(\cdot)$, the objective function is naturally designed as the contrastive loss function, i.e.,

$$\begin{aligned} &\mathrm{y}\,\mathrm{dist}(h_{\boldsymbol{\theta}}(\mathcal{T}(\boldsymbol{z}_i)), h_{\boldsymbol{\theta}}(\mathcal{T}(f(\boldsymbol{Z}))))^2 \\ &\quad + m(1-\mathrm{y})\left[m - \mathrm{dist}(h_{\boldsymbol{\theta}}(\mathcal{T}(\boldsymbol{z}_i)), h_{\boldsymbol{\theta}}(\mathcal{T}(f(\boldsymbol{Z}))))\right]_+^2, \end{aligned} \tag{8}$$

which is minimized with respect to the network parameters $\boldsymbol{\theta}$ via the Adam optimizer of learning rate $1e-4$ [39], where $[x]_+ \triangleq \max\{x, 0\}$, $m := 1$ is a prespecified margin, and $\mathrm{y} \in \{0, 1\}$ is the label; during the training stage, the label is set as $\mathrm{y} := 1$ if the two network inputs are similar (otherwise, $\mathrm{y} := 0$). To see the physical meaning behind the design of (8), note that, when the two inputs are indeed similar (i.e., $\mathrm{y} := 1$), the loss function reduces to $\mathrm{dist}(h_{\boldsymbol{\theta}}(\mathcal{T}(\boldsymbol{z}_i)), h_{\boldsymbol{\theta}}(\mathcal{T}(f(\boldsymbol{Z}))))^2$, meaning that the distance between the two feature vectors

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Algorithm 2 CODE-MM Algorithm for MM

1: **Given** $\boldsymbol{Z} = [z_1, \ldots, z_n]$, and $c > 0$.
2: Compute $f(\boldsymbol{Z})$ using (7).
3: Compute the feature vectors $h_{\boldsymbol{\theta}^\star}(\mathcal{T}(f(\boldsymbol{Z})))$ and $h_{\boldsymbol{\theta}^\star}(\mathcal{T}(z_i))$, $i = 1, \ldots, n$.
4: Compute the distance d_i, $i = 1, \ldots, n$.
5: Compute $\boldsymbol{x}_{\mathrm{DE}}$ for the deep regularization term in the augmented Lagrangian $\mathcal{L}$ of (3).
6: Call Algorithm 1 to compute $\boldsymbol{x}^\star$.
7: **Output** the classification map $\widehat{\boldsymbol{X}} := \mathrm{vec}^{-1}_{n_1 \times n_2}(\boldsymbol{x}^\star)$.

$h_{\boldsymbol{\theta}}(\mathcal{T}(z_i))$ and $h_{\boldsymbol{\theta}}(\mathcal{T}(f(\boldsymbol{Z}))$ should not be large. On the other hand, when the two inputs are dissimilar (i.e., $y := 0$), the loss function reduces to $[m - \mathrm{dist}(h_{\boldsymbol{\theta}}(\mathcal{T}(z_i)), h_{\boldsymbol{\theta}}(\mathcal{T}(f(\boldsymbol{Z}))))]_+^2$, meaning that the distance between the two feature vectors should be large enough (i.e., larger than the prespecified margin m), for which the penalty is zero; otherwise, the smaller the distance in the feature space, the larger the penalty received by the network parameters $\boldsymbol{\theta}$. As alluded in Section III-A, under the CODE framework, small data are sufficient to train the network, leading to the optimized network parameters $\boldsymbol{\theta}^\star$. Specifically, we just use a dataset composed of 70 Sentinel-2 images, each containing 256×256 pixels, where ten images are for validation, while 60 are for training; see Section IV-A for more details about the data preparation.

To complete the implementation of the deep regularizer (2), we just need to report how to obtain $\boldsymbol{x}_{\mathrm{DE}}$ from the trained network $h_{\boldsymbol{\theta}^\star}$. To decide the ith entry of $\boldsymbol{x}_{\mathrm{DE}}$, the input pair of the Siamese network is given by $(\mathcal{T}(f(\boldsymbol{Z})), \mathcal{T}(z_i))$, where zero padding is applied for those boundary pixels z_i when implementing the operator $\mathcal{T}(\cdot)$. Then, the output feature vectors are $h_{\boldsymbol{\theta}^\star}(\mathcal{T}(f(\boldsymbol{Z})))$ and $h_{\boldsymbol{\theta}^\star}(\mathcal{T}(z_i))$, whose distance is denoted as $d_i \triangleq \mathrm{dist}(h_{\boldsymbol{\theta}^\star}(\mathcal{T}(f(\boldsymbol{Z}))), h_{\boldsymbol{\theta}^\star}(\mathcal{T}(z_i)))$. If $d_i < 0.4$, then two center pixels of the two input cubes (i.e., $f(\boldsymbol{Z})$ and z_i) are considered similar, or in other words, z_i is regarded as a pixel similar to the mangrove-like signature $f(\boldsymbol{Z})$. In this case, the ith entry of $\boldsymbol{x}_{\mathrm{DE}}$ is set as 1; otherwise, it is set as 0 [cf. (4)]. Since we just use very small-scaled data, the DE solution $\boldsymbol{x}_{\mathrm{DE}}$ here is just a rough guess, which is, however, sufficient for effective deep REG in our CODE framework, as discussed in Section III-A. This will be further illustrated using experimental evidence next. To conclude, the proposed CODE-MM algorithm is summarized in Algorithm 2 (no further preprocessing stage required), which returns a classification map $\widehat{\boldsymbol{X}} \in [0, 1]^{n_1 \times n_2}$; in case a binary classification map in $\{0, 1\}^{n_1 \times n_2}$ is preferred, a threshold of 0.5 can be set on $\widehat{\boldsymbol{X}}$. The high efficacy of CODE-MM will be experimentally illustrated in Section IV, even if just simple math and small data are used in the CODE theory [37] (cf. Section III-A).

Finally, we illustrate how CODE-MM can be extended to the hyperspectral case. To simplify the illustration, we focus on the hyperspectral data acquired by the Hyperion push broom sensor operated by NASA's EO-1 satellite [7]. The first change (DE part) is to replace the Siamese subnetwork in Fig. 3 with the subnetwork presented in Fig. 4; the reasons behind the replacement are that the high spectral dimensionality of the hyperspectral data limits the spatial size of $\mathcal{T}$ to 5×5

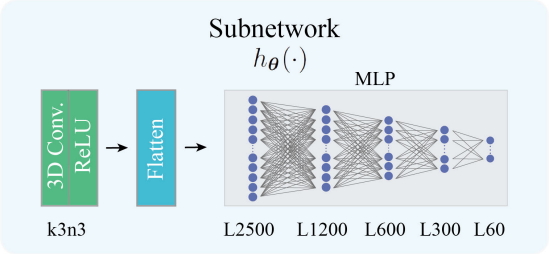


Fig. 4. Subnetwork structure $h_{\boldsymbol{\theta}}(\cdot)$ in the Siamese network for hyperspectral MM, where "k(s)n(c)" means the c channels with kernel size s and "L(r)" means r neurons involved in the corresponding layer.

(implying that the number of valid convolution layers is also limited), and hyperspectral data are more informative comparing to Sentinel-2 data (implying that the number of output MLP features should be higher). It is imperative to emphasize that such a high spectral dimension will inevitably increase the memory burden during the training process; therefore, to strike a balance between the window size w and memory usage, the window size w is adjusted to 5. The second change (CO part) is about the DF term (5), where $\boldsymbol{Z}$ should be replaced by the considered hyperspectral image. The four spectral bands for computing the rough signature $f(\boldsymbol{Z})$ [cf. (6) and (7)] should be replaced by the corresponding Hyperion hyperspectral bands; specifically, the Sentinel-2 bands $(3, 8, 11, 12)$ are replaced by the Hyperion bands $(21, 49, 146, 204)$, respectively. Note that the above selection is to align the central wavelengths of the selected bands between the Sentinel-2 and Hyperion data. Other details about the hyperspectral extension (e.g., small data description and threshold setting) will be presented in Section IV-D. To conclude the section, we remark that, for both Sentinel-2 data and hyperspectral data, the proposed CODE-MM algorithm has a convergence guarantee due to its convexity, as stated in Theorem 1 and proved in Appendix.

Theorem 1: Assume that the DF term and the REG term in the criterion (1) are defined by (5) and (2), respectively. Every limit point of the sequence $\{(\boldsymbol{x}^q, \boldsymbol{z}^q)\}$ generated in the CODE-MM algorithm with $c > 0$ must be an optimal solution of (1). □

IV. Experimental Results and Discussion

In this section, we aim to demonstrate the superiority of the proposed CODE-MM using both quantitative and qualitative comparisons (cf. Section IV-B) based on the experimental settings presented in Section IV-A. Also, in Section IV-C, we show that the setting of the threshold α has less impact on the classification performance of CODE-MM, showing better practical applicability; this, in fact, overcomes a major drawback of the high sensitivity to the threshold setting in other peer methods (e.g., index-based methods). Furthermore, the versatility of the proposed CODE-MM is demonstrated by further applying it to test hyperspectral satellite images in Section IV-D.

A. Data Preparation and Experimental Settings

Mangrove forests, mostly located between the 30°N and 30°S latitudes, are largely restricted to the tropics region. Southeast Asia is the key investigation area accounting for

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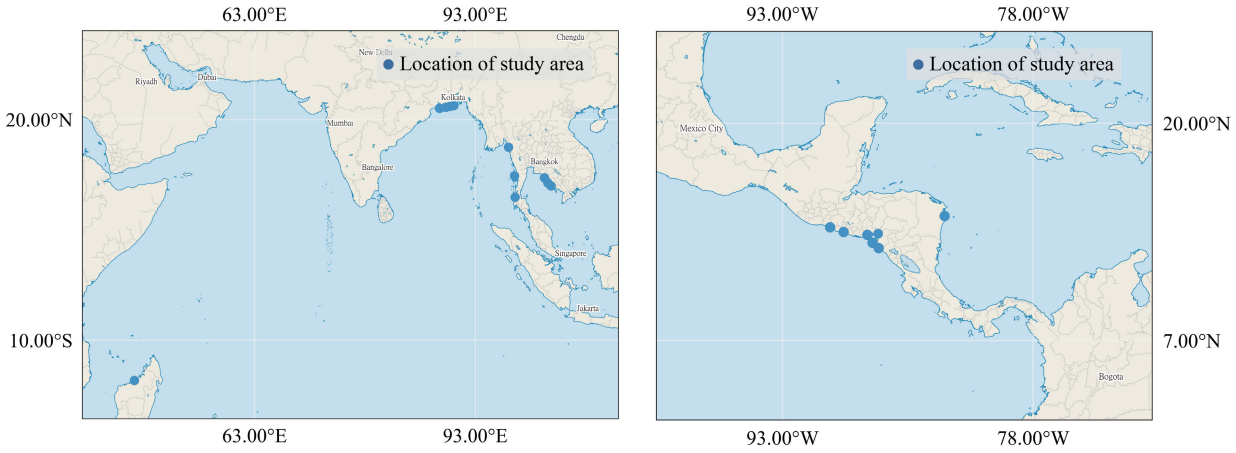


Fig. 5. Locations of the study areas in Southeast Asia (Cambodia, Thailand, and Myanmar), Central America (El Salvador, Nicaragua, and Honduras), South Asia (Bangladesh and India), and Africa (Madagascar).

TABLE III
LOCATION, IMAGE SIZE, AND THE USAGE OF THE SENTINEL-2 DATA USED IN OUR EXPERIMENTS

ID	Year	(Lon, Lat)*	Image Size	Usage
El Salvador-1	2018	(-90.137, 13.643)	1500×2794	Training/Validation
El Salvador-2	2018	(-89.342, 13.118)	4501×12420	Training/Validation
El Salvador-3	2018	(-87.920, 13.320)	2248×2017	Training/Validation
Nicaragua-1	2018	(-87.610, 12.713)	4489×6986	Training/Validation
Nicaragua-2	2018	(-87.401, 12.244)	5228×6210	Training/Validation
Nicaragua-3	2018	(-83.496, 14.397)	6783×4657	Training/Validation
Bangladesh-1	2018	(89.014, 21.673)	6345×4116	Training/Validation
Bangladesh-2	2018	(89.493, 21.687)	6092×4416	Training/Validation
Bangladesh-3	2018	(89.917, 21.769)	3815×10519	Training/Validation
Cambodia-1	2018	(103.310, 10.956)	3788×6058	Testing
Cambodia-2	2018	(102.874, 11.365)	5302×3787	Testing
Thailand-1	2018	(98.337, 9.411)	6045×3778	Testing
Thailand-2	2018	(102.411, 12.058)	3022×3022	Testing
Myanmar-1	2018	(98.346, 12.316)	3045×1980	Testing
Myanmar-2	2018	(97.347, 16.175)	4265×3807	Testing
India	2018	(88.758, 21.645)	1656×1442	Testing
Honduras	2018	(-87.714, 13.266)	3272×4657	Testing
Thailand	2018	(102.411, 12.058)	256×256	Sensitivity Study
Madagascar	2019	(46.687, -15.449)	256×256	Hyperspectral Case

* denotes the location of the bottom-left corner pixel of the image.

more than one-third of the global mangrove forest extent with the greatest mangrove species diversity [55]. Table III summarizes the location (in terms of the longitude (Lon) and latitude (Lat) of the bottom-left corner pixel), retrieval year, and image size of Sentinel-2 data for the experimental investigation. The study areas include three regions in Southeast Asia, including Cambodia, Thailand, and Myanmar. Moreover, El Salvador, Nicaragua, and Honduras in Central America, Bangladesh and India in South Asia, and Madagascar in Africa are also included, as depicted in Fig. 5. The datasets are applied for model training/validation and testing (i.e., quantitative/qualitative comparison study) in Section IV-B, as well as various experiments for the sensitivity study in Section IV-C. It is crucial to emphasize that the nations of six testing data are distinct from the nations of 180 training and 20 validation data, and this non-overlapping setting is to verify whether the studied methods are broadly applicable or limited to some specific countries. The datasets also serve as a baseline comparison for the hyperspectral experiment in Section IV-D.

All these Sentinel-2 datasets, ranging from 443 to 2280 nm, were acquired over 2018–2019 and downloaded from the GEE [56]. One of the most difficult parts is to obtain the corresponding ground-truth classification maps for training/validation/testing purposes. Fortunately, Clark Labs [57] has provided this crucial information for the region of interest (ROI) under investigation, besides many other services, including habitat assessment, species distribution modeling, and evaluation of ecosystem services. Specifically, Clark Labs [57] has a product for the project "Coastal Habitat Mapping: Mangrove and Pond Aquaculture Conversion," where land cover maps of three classes (i.e., mangrove, coastal wetland, and pond aquaculture) are provided over ten countries in 2018. The first land cover class (i.e., mangrove) is sifted out to serve as the ground-truth classification maps of 2018, which are then manually adjusted based on visual inspections to obtain the final ground-truth classification maps. Furthermore, the Sentinel-2 images downloaded from GEE are of coarser resolution and are preprocessed by the SSSS software [10] to uniformly achieve the highest 10-m resolution for all the spectral bands.

During the training stage, the Adam optimizer adopts a batch size of $5e + 4$ with the number of epochs set as 80; note that training pairs are composed of small-sized subtensor of dimension $9 \times 9 \times 12$ (rather than the complete tensor $\boldsymbol{Z}$) as discussed previously, allowing such a large batch size. The training loss employs contrastive loss with the learning rate set as $1e - 4$, as elaborated in Section III-E. Subsequently, the learning weight for testing is determined based on the optimal validation loss achieved during the training phase. For the training phase, the input pair is formed by $\mathcal{T}(\boldsymbol{z}_i)$ and $\mathcal{T}(\boldsymbol{z}_j)$, with the corresponding label y [cf. (8)] set based on whether the center pixels $\boldsymbol{z}_i$ and $\boldsymbol{z}_j$ belong to the same category; whether they belong to the same category or not can be automatically justified based on the ground-truth information discussed above. In the Siamese network, the setting of input pairs during the testing stage has been presented in

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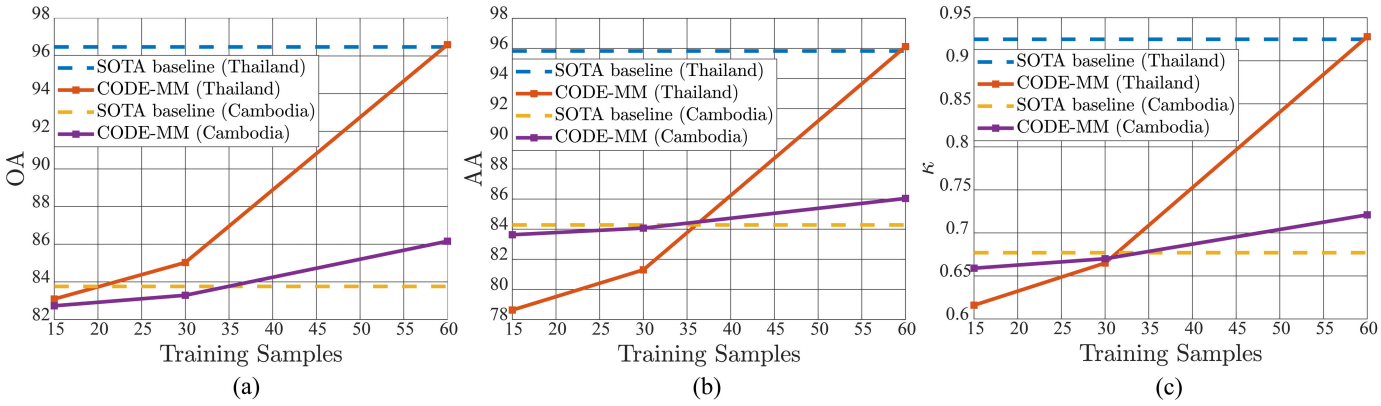


Fig. 6. Performance assessment under different numbers of training samples compared to the SOTA baseline (i.e., the strongest peer method), where the Thailand data exhibit a more continuous mangrove distribution (cf. Fig. 7) with the SOTA baseline being GC-UNet (cf. Table IV), whereas the Cambodia data show relatively complicated textures of interspersed non-mangrove regions (cf. Fig. 8) with the SOTA baseline being Capsules-UNet (cf. Table V). For both representative scenarios, one can see that 60 is a safe number for CODE-MM to outperform the baseline methods. (a) OA, (b) AA, and (c) κ.

Section III-E. Both the training and testing stages are executed on Python 3.8.8 running on a desktop computer equipped with NVIDIA RTX-3090 GPU and with Core-i9-10900X CPU (3.70-GHz speed and 64-GB RAM).

B. Quantitative and Qualitative Analyses

In this section, we compare the proposed CODE-MM with existing benchmark classification methods, including NDVI [58], NDWI [17], SAVI [16], MMRI [14], MVI [9], UNet [22], SegNet [23], GC-UNet [25], DCNN [26], ME-Net [27], and Capsules-UNet [29]. To have a fair comparison for all the deep learning-based peer methods (i.e., UNet, SegNet, GC-UNet, DCNN, ME-Net, and Capsules-UNet), the learning procedure is conducted using 180 training images and 20 validation images, with a learning rate of $1e-4$, a batch size of 6, and epochs of 100. To verify the efficacy of the proposed method for learning with limited data, only 60 training images and 20 validation images are adopted for CODE-MM. As the success of deep learning techniques is heavily reliant on the amount of available data, a straightforward proof to demonstrate the benefits of our small-data learning technique is that CODE-MM has achieved superior performance using only 60 limited training samples by saving 67% of data usage required for training other SOTA deep learning methods, including UNet, SegNet, GC-UNet, DCNN, ME-Net, and Capsules-UNet. Besides, extensive experiments will be conducted to examine the efficacy of the proposed small-data learning method using various data across Central America, South Asia, Southeast Asia, and Africa, demonstrating the universal applicability of CODE-MM, as will be discussed in Tables IV–VIII and XII. Furthermore, we have conducted an ablation study using different training dataset sizes to assess the impact of CODE-MM and found that using 60 training samples is indeed a suitable default setting, as demonstrated in Fig. 6. One can also see that 60 training samples seem to be the limit of the current CODE theory, as indicated by Fig. 6.

The threshold value of NDVI is set as 0.33, which is the default setting for the case of moderate mangrove density [59]. The default threshold value of NDWI is set as -0.15 [60]. As for SAVI, MMRI, and MVI, we have tried our best to tune their threshold values as 0.27, -0.27, and 2.6, respectively, to optimize their qualitative and quantitative performances. For SAVI, there is another parameter, i.e., the soil-adjustment factor L, whose default value for intermediate vegetation density is set as 0.5 [16]. Also, the REG parameter in CODE-MM is set as $\lambda := 0.5$. Accordingly, the experimental results are summarized in Table IV and Fig. 7 for the Thailand data, Table V and Fig. 8 for the Cambodia data, and Table VI and Fig. 9 for the Myanmar data. In addition, Table VII and Fig. 10 are for the India data, while Table VIII and Fig. 11 are for the Honduras data.

TABLE IV
QUANTITATIVE ANALYSIS USING THAILAND DATA

Method	OA (%)	AA (%)	κ
NDVI	76.27	69.30	0.435
NDWI	70.53	61.88	0.277
SAVI	81.96	76.86	0.585
MMRI	92.91	92.00	0.849
MVI	93.66	92.99	0.865
UNet	93.14	91.26	0.851
SegNet	88.88	85.72	0.753
GC-UNet	96.48	95.83	0.925
DCNN	94.86	94.73	0.892
ME-Net	95.03	94.16	0.894
Capsules-UNet	94.87	93.83	0.890
CODE-MM	**96.60**	**96.12**	**0.928**

TABLE V
QUANTITATIVE ANALYSIS USING CAMBODIA DATA

Method	OA (%)	AA (%)	κ
NDVI	54.35	57.39	0.139
NDWI	47.92	51.39	0.026
SAVI	71.14	71.93	0.431
MMRI	74.85	75.56	0.503
MVI	73.10	73.38	0.464
UNet	75.02	75.95	0.508
SegNet	74.29	75.86	0.500
GC-UNet	74.72	75.94	0.505
DCNN	78.96	78.43	0.573
ME-Net	73.47	75.13	0.485
Capsules-UNet	83.76	84.28	0.677
CODE-MM	**86.16**	**86.04**	**0.721**

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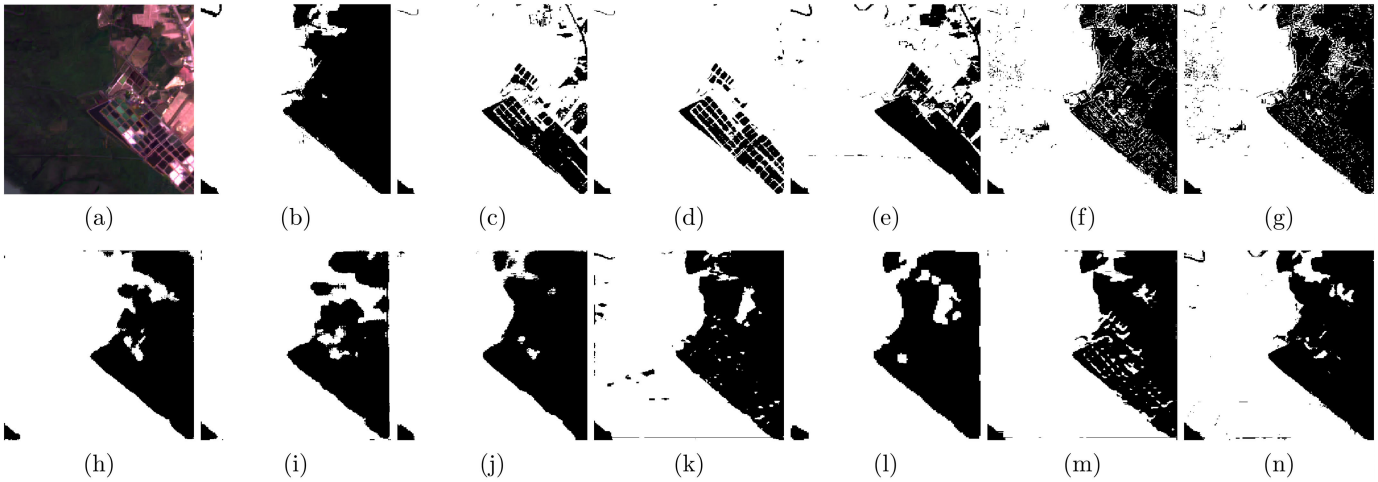


Fig. 7. Qualitative study using Thailand data. (a) RGB reference. (b) Ground-truth map. Classification maps obtained by (c) NDVI, (d) NDWI, (e) SAVI, (f) MMRI, (g) MVI, (h) UNet, (i) SegNet, (j) GC-UNet, (k) DCNN, (l) ME-Net, (m) Capsules-UNet, and (n) proposed CODE-MM.

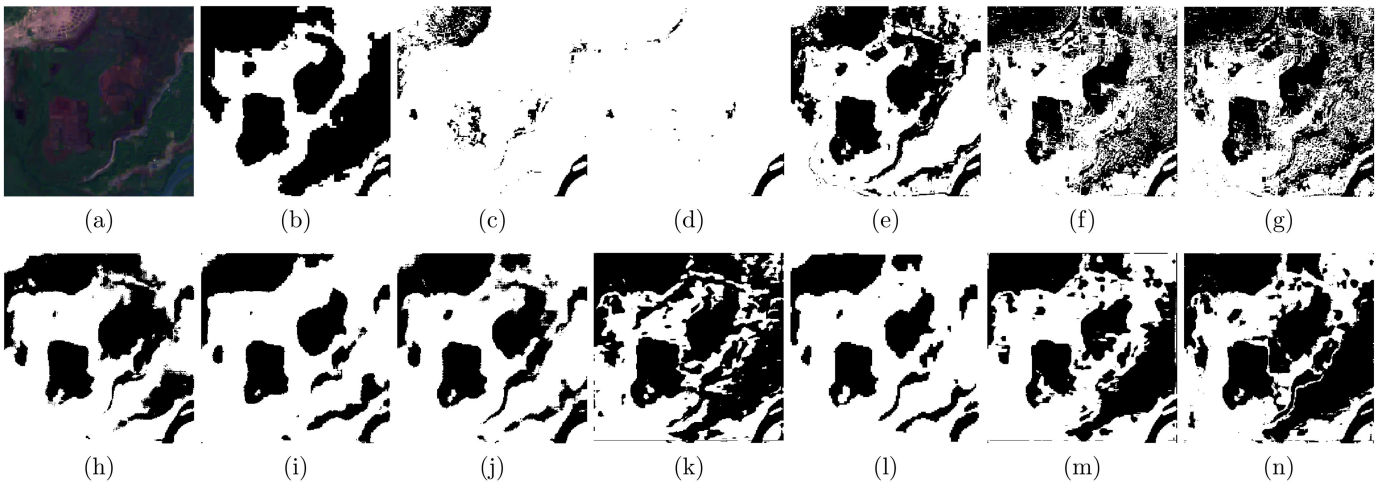


Fig. 8. Qualitative study using Cambodia data. (a) RGB reference. (b) Ground-truth map. Classification maps obtained by (c) NDVI, (d) NDWI, (e) SAVI, (f) MMRI, (g) MVI, (h) UNet, (i) SegNet, (j) GC-UNet, (k) DCNN, (l) ME-Net, (m) Capsules-UNet, and (n) proposed CODE-MM.

Quantitative performance indices used in Tables IV–VIII are explicitly defined in the following for rigorousness. In order to rigorously define the three commonly used classification performance indices, including overall accuracy (OA) [61], average accuracy (AA) [62], and kappa statistics (κ) [63], let P_{ij} be the probability of the classifier returning class i when the ground truth is actually class j.

1) The OA index is to compute the proportion of correctly classified samples/pixels, i.e.,

$$\mathrm{OA} \triangleq \sum_{n=1}^{N} P_{nn},$$

where $N = 2$ is the number of total classes.

2) The AA index is to compute the averaged conditional probability of correct classification given each class, i.e.,

$$\mathrm{AA} = \frac{1}{N} \sum_{n=1}^{N} \frac{P_{nn}}{P_{n}},$$

where $P_{n.} = \sum_{n'=1}^{N} P_{nn'}$.

3) The kappa statistics is defined as

$$\kappa = \frac{\mathrm{OA} - p_e}{1 - p_e},$$

where p_e is the probability of a "random-guessing algorithm" yielding correct classification results, i.e.,

$$p_e = P_{1.}P_{.1} + P_{2.}P_{.2} + \cdots + P_{N.}P_{.N},$$

in which $P_{.n} = \sum_{n'=1}^{N} P_{n'n}$.

The RGB reference and the ground-truth map for the Thailand data are displayed in Fig. 7(a) and (b). Classification maps obtained by NDVI, NDWI, SAVI, MMRI, MVI, UNet, SegNet, GC-UNet, DCNN, ME-Net, Capsules-UNet, and the proposed CODE-MM are shown in Fig. 7(c)–(n), respectively; pixels with white color (resp., black color) indicate the mangrove region (resp., non-mangrove region). One can observe that only the classification maps returned by GC-UNet and the proposed CODE-MM hold a good resemblance to the ground-truth map presented in Fig. 7(b). However, it has been observed that some pixels belonging to mangroves are

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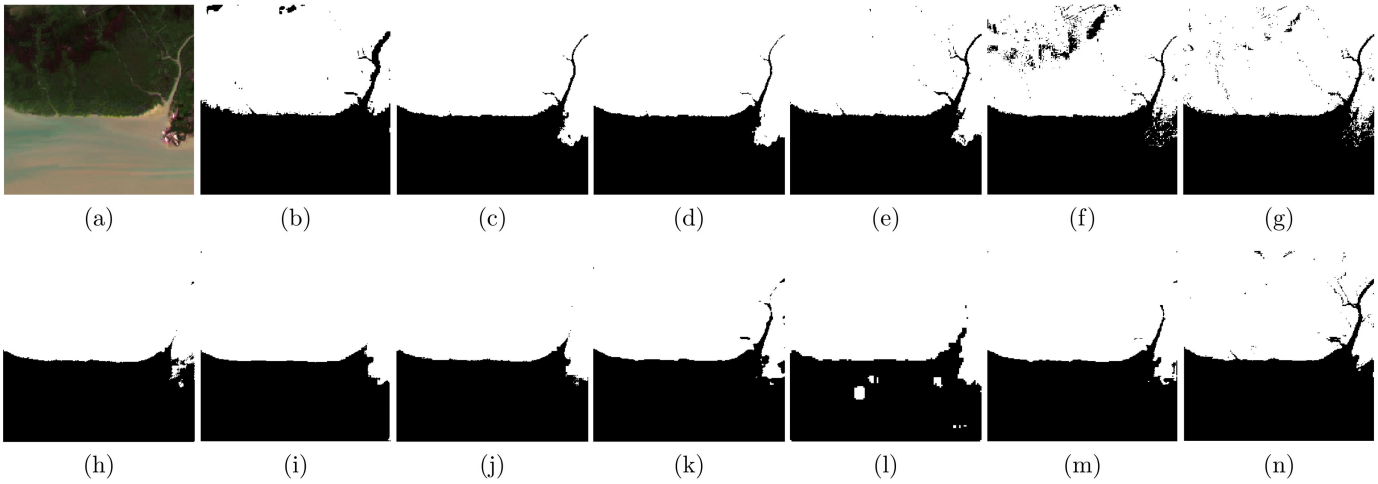


Fig. 9. Qualitative study using Myanmar data. (a) RGB reference. (b) Ground-truth map. Classification maps obtained by (c) NDVI, (d) NDWI, (e) SAVI, (f) MMRI, (g) MVI, (h) UNet, (i) SegNet, (j) GC-UNet, (k) DCNN, (l) ME-Net, (m) Capsules-UNet, and (n) proposed CODE-MM.

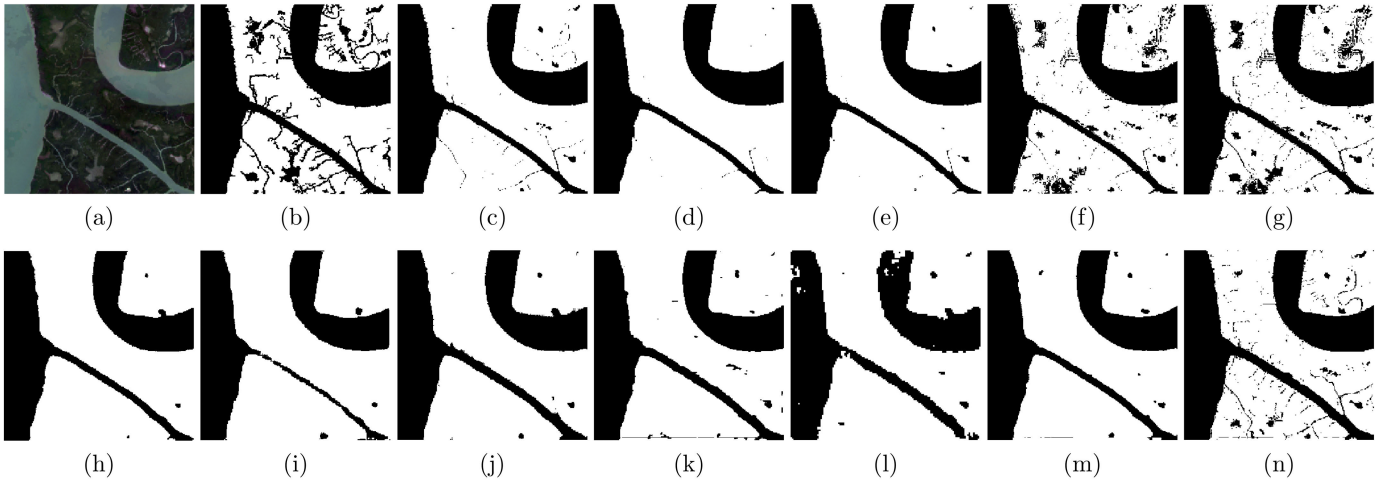


Fig. 10. Qualitative study using India data. (a) RGB reference. (b) Ground-truth map. Classification maps obtained by (c) NDVI, (d) NDWI, (e) SAVI, (f) MMRI, (g) MVI, (h) UNet, (i) SegNet, (j) GC-UNet, (k) DCNN, (l) ME-Net, (m) Capsules-UNet, and (n) proposed CODE-MM.

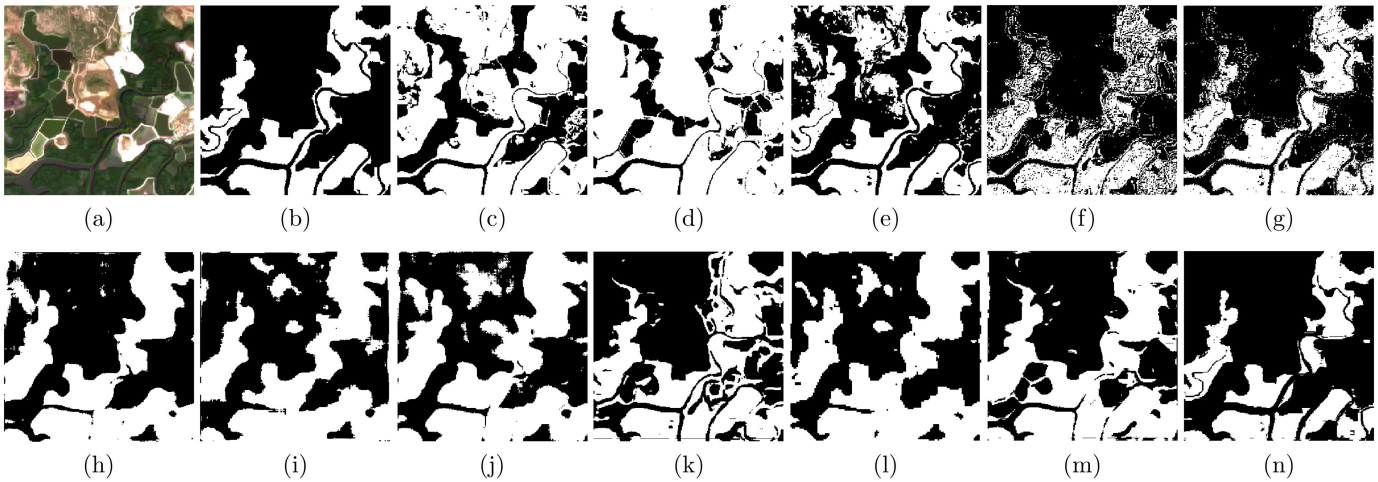


Fig. 11. Qualitative study using Honduras data. (a) RGB reference. (b) Ground-truth map. Classification maps obtained by (c) NDVI, (d) NDWI, (e) SAVI, (f) MMRI, (g) MVI, (h) UNet, (i) SegNet, (j) GC-UNet, (k) DCNN, (l) ME-Net, (m) Capsules-UNet, and (n) proposed CODE-MM.

misclassified in the upper-left region of the GC-UNet classification map, as depicted in Fig. 7(j). Thus, the proposed CODE-MM has the best qualitative performance in this case study. This well coincides with the quantitative analysis presented in Table IV, where boldfaced numbers are to indicate the best performance with respect to a particular performance

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TABLE VI
QUANTITATIVE ANALYSIS USING MYANMAR DATA

Method	OA (%)	AA (%)	κ
NDVI	95.18	94.75	0.902
NDWI	94.78	94.31	0.894
SAVI	95.91	95.57	0.917
MMRI	94.06	94.00	0.880
MVI	96.55	96.37	0.930
UNet	96.90	96.64	0.937
SegNet	95.86	95.50	0.916
GC-UNet	96.46	96.16	0.928
DCNN	96.96	96.75	0.939
ME-Net	95.48	95.13	0.908
Capsules-UNet	96.72	96.45	0.934
CODE-MM	**97.65**	**97.58**	**0.953**

TABLE VII
QUANTITATIVE ANALYSIS USING INDIA DATA

Method	OA (%)	AA (%)	κ
NDVI	90.12	88.86	0.795
NDWI	89.45	88.03	0.780
SAVI	89.29	87.87	0.777
MMRI	87.63	86.80	0.746
MVI	89.25	88.70	0.780
UNet	89.14	87.88	0.775
SegNet	87.80	86.24	0.745
GC-UNet	89.49	88.40	0.783
DCNN	88.53	87.41	0.763
ME-Net	88.25	87.15	0.757
Capsules-UNet	88.69	87.42	0.765
CODE-MM	**90.31**	**89.58**	**0.801**

TABLE VIII
QUANTITATIVE ANALYSIS USING HONDURAS DATA

Method	OA (%)	AA (%)	κ
NDVI	68.00	75.61	0.418
NDWI	54.10	65.15	0.229
SAVI	85.19	88.13	0.698
MMRI	83.22	81.25	0.626
MVI	91.44	91.12	0.812
UNet	86.92	89.28	0.729
SegNet	84.27	86.75	0.677
GC-UNet	79.82	83.93	0.602
DCNN	80.97	82.42	0.605
ME-Net	82.60	85.99	0.650
Capsules-UNet	84.46	87.33	0.683
CODE-MM	**93.77**	**92.06**	**0.858**

index. CODE-MM achieves SOTA performances for all the performance measures for the Thailand data. This alludes that introducing spatial continuity into the deep regularizer design is indeed a promising approach under the CODE framework.

Next, we study a highly challenging scenario using the Cambodia data that involves delicate textures, as can be seen from Fig. 8(a) and (b). All the methods under investigation show degraded quantitative performances in Table V in such complicated terrain. From Fig. 8, those index-based methods and DE-based methods cannot recover the fine details indicated in the ground-truth map, suffer from misclassifying considerable non-mangrove regions, and contain small black cavities, while the proposed CODE-MM has rather strong qualitative performance as also indicated by its highest kappa statistic value in Table V. Compared with the Thailand/Cambodia datasets, the land cover of Myanmar is simpler, as shown in Fig. 9, where CODE-MM has the best qualitative performances, especially in the upper-right part of the classification map. From Table VI, the proposed CODE-MM has achieved the best quantitative performance. From Fig. 10, it becomes apparent that identifying the small tributary in Fig. 10(a) and (b) is challenging. Notably, NDVI, MMRI, MVI, and the proposed CODE-MM can efficiently identify complex characteristics of the tributary, particularly within the lower region, as displayed in Fig. 10(c), (f), (g), and (n). Another intricate terrain with small tributaries is displayed in Fig. 11(a) and (b). Fig. 11(n) demonstrates that COME-MM is capable of accurately classifying both texture and shape. On the other hand, other peer methods exhibit deficiencies, such as the misclassification of large regions or existing evident disturbances in Fig. 11(c)–11(m). Besides, the limitations of index-based methods are apparent, as presented in Figs. 7–11. As one can see, these methods fail to accurately represent the spatial continuity or the structure of mangrove formations when classifying individual mangrove pixels separately. Although those DE-based methods may provide more precise classification results than index-based methods, their results still suffer from misclassifying non-mangrove regions. In contrast, our novel approach accounts for spatial continuity, resulting in the preservation of the mangrove shape with minimal misclassification, which is reflected in the superior performance in Tables IV–VIII.

Finally, the computational time of studied methods averaged over the five datasets is presented in Table IX. It is not surprising that those index-based methods are very fast because they just perform pixel-by-pixel computation using simple formulae without algorithmically iterative procedure; cf. Table I. However, all these classification methods under investigation are fast enough from the view of practical applicability. To conclude this section, at the cost of computational complexity, the proposed CODE-MM developed under the ADMM-Adam optimization, combining the advantages of CO and DE, achieves SOTA classification performances, especially when the land covers involve complicated textures. Further improving the computational efficiency is also a good future research line.

C. Sensitivity Study for the Threshold Setting

To automatically compute the multispectral signature $f(\boldsymbol{Z})$ in (7) for the subsequent deep REG task, we need to properly set a threshold value α when estimating the mangrove area $\mathcal{M}$. Unlike other peer methods that are highly sensitive to the setting of the threshold values (index-based methods, in particular), this experiment aims at demonstrating the great superiority of CODE-MM; specifically, we will show that the performance of CODE-MM is much less sensitive to the threshold setting. From Section IV-B, MVI is regarded as the strongest peer method among all index-based methods, so it is adopted to serve as the baseline in this sensitivity study.

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TABLE IX
COMPUTATIONAL TIME [MEASURED IN SECONDS (SEC.)] TAKEN BY VARIOUS CLASSIFICATION METHODS UNDER INVESTIGATION

Method	NDVI	NDWI	SAVI	MMRI	MVI	UNet	SegNet	GC-UNet	DCNN	ME-Net	Capsules-UNet	CODE-MM
Time (sec.)	**0.0008**	**0.0008**	**0.0008**	0.0014	**0.0008**	0.4668	0.2955	1.2986	0.2854	0.8813	1.1413	0.6038

TABLE X
KAPPA STATISTICS OF THE CLASSIFICATION RESULTS RETURNED BY MVI AND THE PROPOSED CODE-MM OVER A WIDE RANGE OF THRESHOLD VALUES α

Method	Threshold: 1	Threshold: 2	Threshold: 3	Threshold: 4	Threshold: 5	Threshold: 6	Threshold: 7	Threshold: 8
MVI	0.4448	0.6479	0.5622	0.1698	-0.0048	-0.0205	-0.0209	-0.0191
CODE-MM	0.8644	0.8639	0.8640	0.8637	0.8634	0.8633	0.8639	0.8638

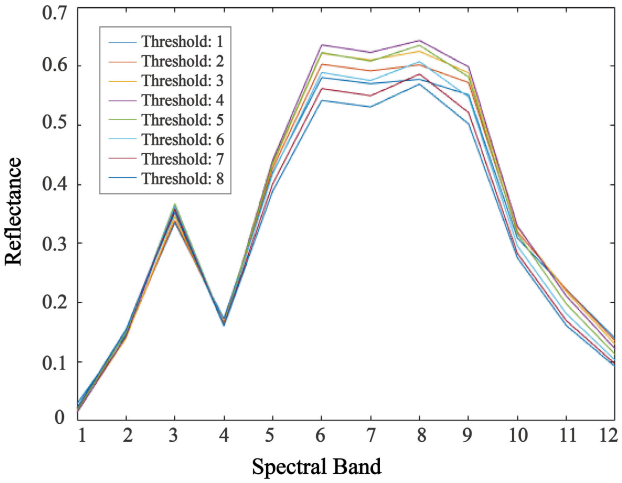


Fig. 12. Mangrove multispectral signatures $f(\boldsymbol{Z})$ estimated using different threshold values α.

In Table X, we display the kappa statistics of the classification results yielded by MVI and CODE-MM over an extensive range of threshold values $\alpha \in \{1, \ldots, 8\}$. As one can see, MVI is highly sensitive to the threshold setting because the values of κ vary dramatically for $1 \leq \alpha \leq 4$; and κ even goes to the negative range for $5 \leq \alpha \leq 8$. By contrast, even if α goes through an extensive range, the proposed CODE-MM is highly robust as the kappa statistics keep in a very narrow range of $[0.8633, 0.8644]$, as can be seen from Table X.

We surprisingly found that such an outstanding property comes from the high similarity among the sifted mangrove multispectral signatures. This is illustrated using Fig. 12, where the multispectral signatures $f(\boldsymbol{Z})$ computed via (7) using different values of $\alpha \in \{1, \ldots, 8\}$ hold high resemblance. There are two consequences. First, by (5), the DF term DF$(\cdot)$ remains almost unchanged. Second, by (2) and the definition of $\boldsymbol{x}_{\text{DE}}$ reported at the end of Section III-E, we know that the regularizer REG$(\cdot)$ also remains almost the same. With that being said, the final learning criterion (1) of the CODE theory is almost insensitive to the setting of the threshold value. Since the algorithm used to implement the learning criterion is also irrelevant to α, the resultant kappa statistics are, hence, not surprisingly highly uniform across a wide range of α. Thus, we simply set $\alpha := 6$ for all the Sentinel-2 experiments throughout this article.

TABLE XI
QUANTITATIVE ANALYSIS OF HYPERION HYPERSPECTRAL SATELLITE IMAGERY FROM NASA OVER MADAGASCAR, AFRICA

Method	OA (%)	AA (%)	κ
EMVI	86.37	86.23	0.726
MPVI	90.39	90.54	0.808
NDWVI	86.67	86.77	0.734
UNet	90.99	91.06	0.820
SegNet	85.61	85.44	0.711
GC-UNet	89.88	90.20	0.799
DCNN	80.32	79.84	0.602
ME-Net	88.05	87.92	0.760
Capsules-UNet	90.71	90.89	0.815
CODE-MM	**92.19**	**92.09**	**0.843**

TABLE XII
QUANTITATIVE ANALYSIS OF SENTINEL-2 MULTISPECTRAL SATELLITE IMAGERY FROM ESA OVER MADAGASCAR, AFRICA

Method	OA (%)	AA (%)	κ
NDVI	91.42	91.84	0.829
NDWI	91.10	91.54	0.823
SAVI	89.96	89.84	0.798
MMRI	74.39	73.71	0.480
MVI	71.70	70.62	0.421
UNet	90.51	90.95	0.811
SegNet	89.77	90.22	0.797
GC-UNet	91.80	92.10	0.837
DCNN	88.11	88.42	0.763
ME-Net	80.56	79.88	0.605
Capsules-UNet	88.09	88.67	0.764
CODE-MM	**93.52**	**93.37**	**0.870**

D. Application to Hyperspectral Mangrove Mapping

This section applies the CODE-MM algorithm to achieve hyperspectral MM. To have a baseline comparison, we test CODE-MM not only on Hyperion hyperspectral images but also on Sentinel-2 images, both covering the same ROI of Madagascar, as depicted in Figs. 13(a) and 14(a). Prior to the evaluation stage, we remove the corrupted bands 1–7, 58–76, 121–126, 167–180, and 222–242 from the real hyperspectral data obtained by the Hyperion push broom sensor operated by NASA's EO-1 satellite [7], ranging from 0.4 to 2.5 μm in 242 bands with a 30-m spatial resolution. As the preprocessing stage, the remaining 175 bands then undergo the ENVI

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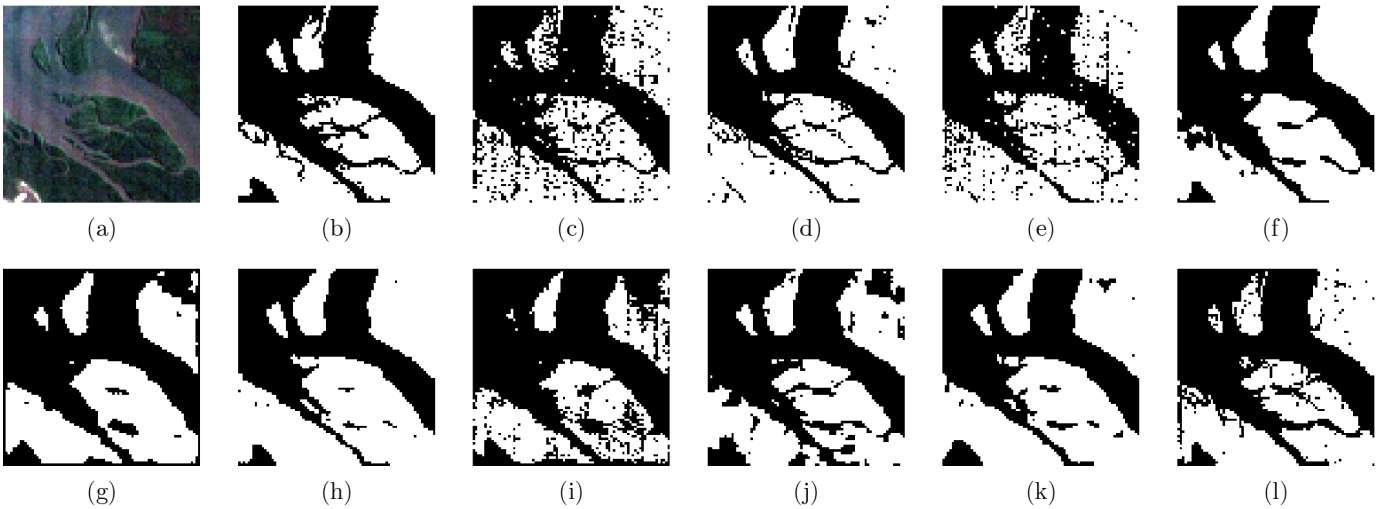


Fig. 13. Qualitative study of Hyperion hyperspectral satellite imagery from NASA over Madagascar, Africa, in 2016. (a) RGB reference. (b) Ground-truth map. Classification maps obtained by (c) EMVI, (d) MPVI, (e) NDWVI, (f) UNet, (g) SegNet, (h) GC-UNet, (i) DCNN, (j) ME-Net, (k) Capsules-UNet, and (l) proposed CODE-MM.

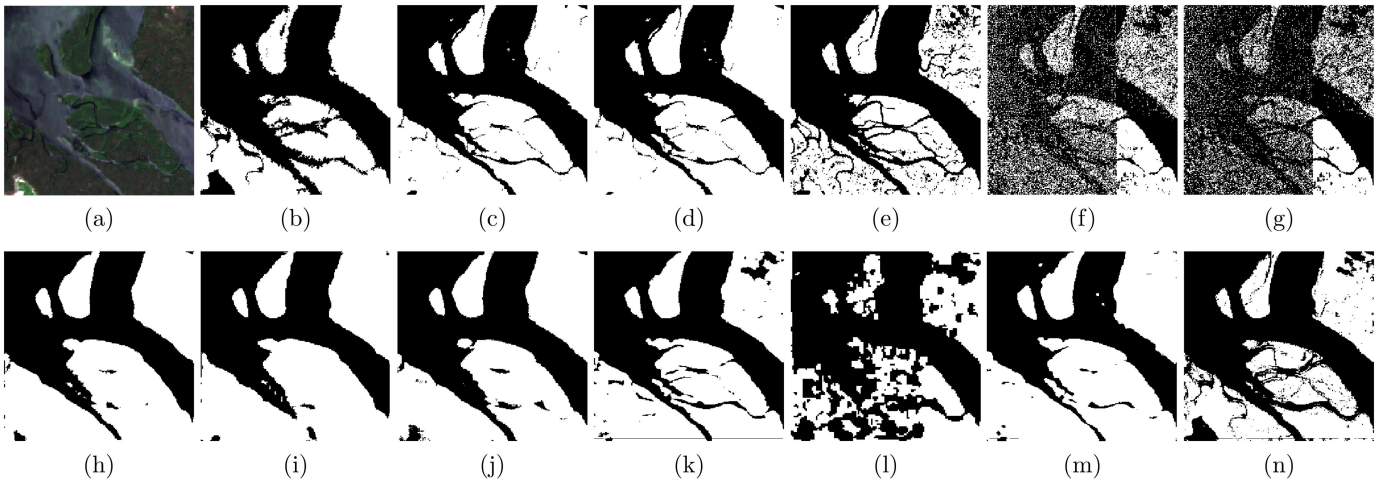


Fig. 14. This set of Sentinel-2 experiments mainly serves as the baseline for the hyperspectral test. Qualitative study of Sentinel-2 multispectral satellite imagery from ESA over Madagascar, Africa, in 2019. (a) RGB reference. (b) Ground-truth map. Classification maps obtained by (c) NDVI, (d) NDWI, (e) SAVI, (f) MMRI, (g) MVI, (h) UNet, (i) SegNet, (j) GC-UNet, (k) DCNN, (l) ME-Net, (m) Capsules-UNet, and (n) proposed CODE-MM.

radiometric calibration and the fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASH) atmospheric correction [64]. Moreover, the ground-truth labeling is done using manual adjustments and visual inspections, resulting in the ground-truth maps presented in Figs. 13(b) and 14(b), which are for Hyperion (2016) and Sentinel-2 (2019) satellites, respectively.

In order to investigate the applicability of the proposed CODE-MM framework on hyperspectral images, we compare CODE-MM with various benchmark hyperspectral MM methods, including the enhanced MVI (EMVI), mangrove probability VI (MPVI), and normalized difference wetland VI (NDWVI), as reported in previous studies [65], [66]. Besides, various multispectral DE methods are utilized for verifying the efficacy of the proposed CODE-MM, including UNet [22], SegNet [23], GC-UNet [25], DCNN [26], ME-Net [27], and Capsules-UNet [29]. The hyperspectral index-based methods are briefly introduced in the following.

1) EMVI effectively distinguishes mangroves from terrestrial vegetation through two empirical observations. First, the green reflectance value of mangroves is usually lower than that of other terrestrial vegetation. Second, the SWIR1 and SWIR2 reflectance values of mangroves are significantly lower than that of other vegetation. Therefore, EMVI is expressed as

$$\text{EMVI} = \frac{\text{Green} - \text{SWIR2}}{\text{SWIR1} - \text{Green}},$$

where Green, SWIR1, and SWIR2 are the reflectance values in bands 21, 146, and 204 with central wavelengths of 559, 1608, and 2193 nm, respectively.

2) MPVI determines the similarity between the spectrum of the target pixel and the mangrove candidate pixel by calculating their spectral correlation fitting [67] to discriminate whether it belongs to the mangrove or not, expressed as

$$\frac{n\sum_{i=1}^{n} \boldsymbol{R}_i \boldsymbol{r}_i - \sum_{i=1}^{n} \boldsymbol{R}_i \sum_{i=1}^{n} \boldsymbol{r}_i}{\sqrt{n\sum_{i=1}^{n} \boldsymbol{R}_i^2 - (\sum_{i=1}^{n} \boldsymbol{R}_i)^2}\sqrt{n\sum_{i=1}^{n} \boldsymbol{r}_i^2 - (\sum_{i=1}^{n} \boldsymbol{r}_i)^2}},$$

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where n is the total number of the spectral bands in the image, $\boldsymbol{R}_i$ is the reflectance value at band i of the target pixel, and $\boldsymbol{r}_i$ is the value at band i of the mangrove candidate spectrum. Here, the mangrove candidate spectrum $\boldsymbol{r}$ can be automatically inferred from the testing image itself by using $f(\boldsymbol{Z})$ [cf. (7)].

3) Since the habitat of mangrove forests is mostly in the coastal and wetland regions with lower temperatures and higher moisture, their spectral reflectances in the SWIR band are lower than that of terrestrial vegetation. NDWVI considers the green band (ranging from 528 to 589 nm) and the SWIR band (ranging from 2113 to 2284 nm) to distinguish mangrove forests from terrestrial vegetation, defined as

$$\text{NDWVI} = \frac{\text{SWIR} - \text{Green}}{\text{SWIR} + \text{Green}},$$

where Green and SWIR are the average reflectance values over the green and SWIR bands, respectively.

In the training phase, only 18 Hyperion images with a size of 256×256 are utilized for training the hyperspectral Siamese network, while two images are for validation. To apply the proposed CODE-MM on hyperspectral images in the Siamese network, the window size is set to $w := 5$, and the subnetwork structure is modified, as shown in Fig. 4. The optimization is carried out using the Adam optimizer with the learning rate of $1e-3$, and the threshold value of the mangrove signature $f(\boldsymbol{Z})$ is set to $\alpha := 10$. Accounting for the energy variability that arises from differences in spectral numbers between multispectral and hyperspectral images, for hyperspectral images, we adjust the threshold on classification map $\widehat{\boldsymbol{X}}$ to be half of the maximum value after removing a quarter of outliers.

For evaluating the Hyperion image over Madagascar containing $n = 83 \times 83$ pixels, we also adjust the thresholds for EMVI, MPVI, and NDWVI to generate the best qualitative and quantitative results; the threshold of EMVI is -0.3, and the upper and lower thresholds of MPVI are 0.72 and 1.4, while the upper and lower thresholds of NDWVI are -0.55 and -0.15, respectively. As illustrated in Fig. 13, the classification maps generated by EMVI, MPVI, and NDWVI are noisy between the mangrove and river in the inland region. However, the proposed CODE-MM accurately depicts the mangrove areas, separates the river from the mangrove quite effectively, and outperforms other peer methods as evidenced by all metrics in Table XI.

In Fig. 14, for baseline comparison, exactly the same ROI on Sentinel-2 data containing $n = 256 \times 256$ pixels is evaluated among all studied methods investigated in Section IV-B, where the thresholds of index-based methods are set the same as described in Section IV-B. Apparently, most of the peer methods misclassify the pixels in the inland region; in contrast, the classification map obtained from the proposed CODE-MM effectively depicts the mangrove areas, as shown in Fig. 14(n), whose excellent performance can also be seen in Table XII. Therefore, the proposed CODE-MM well demonstrates its outstanding classification performance on both multispectral and hyperspectral images, showing its great potential in accurate MM.

V. Conclusion

For effectively addressing the critical MM problem, merely using index-based methods is insufficient to well exploit the spatial continuity nature of the mangrove distribution, resulting in scattered artifacts in the final classification map. Deep learning (DE) methods have been experimentally shown to be of good potential to incorporate such a spatial property in improving mapping accuracy. However, DE-based methods require considerable human resources for big data collection and labeling, making it a less practical technique. This motivated us to further incorporate CO into DE for achieving effective small-data learning [37], thereby leading to the proposed CODE-MM algorithm (i.e., Algorithm 2).

CODE-MM is first developed for the Sentinel-2 satellite (the mainstream satellite for the MM mission) by judiciously blending a spatial–spectral-aware Siamese deep regularizer and a customized convex analysis technique (i.e., relaxation-based ADMM) and subsequently generalized for addressing the hyperspectral satellite case. As it turns out, CODE-MM yields SOTA MM performances on both the multispectral and hyperspectral experiments. Though further taking the spatial information into account brings additional computing costs compared to other pixelwise index-based methods, CODE-MM is fast enough for practical application, and such computational efficiency is achieved by the derived closed-form solutions of all the algorithmic steps.

Unlike most benchmark MM methods that are quite sensitive to the threshold setting, the performance of CODE-MM is insensitive to that setting, yielding uniformly good performances over a wide range of threshold values (cf. Section IV-C). Moreover, some benchmark MM methods (e.g., MPVI) require the users to manually select some reference mangrove signatures, but this is not very convenient in practice; in CODE-MM, however, we have proposed a mangrove signature estimation method [cf. (7)], which can be used to automatically and adaptively infer the desired reference signature directly from the testing image.

Appendix
Proof of Theorem 1

One can easily verify that the $\boldsymbol{Q}$-quadratic norm in (2) is a convex function. Also, (5) can be viewed as a composition of a convex function and an affine function, implying that (5) is also a convex function [38]. Since the domain $\mathcal{S}$ is a convex set after adopting the convex relaxation trick (cf. Section III-C), the criterion (1) is a convex problem with $\boldsymbol{x}$ and $\boldsymbol{z}$ in Algorithm 1 being linearly associated by the identity matrix, i.e., $\boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}$ with $\boldsymbol{A} = \boldsymbol{I}$. Then, as long as $c > 0$, the CO theory [38, Lemma 9.2] implies that the invertibility of $\boldsymbol{A}^T\boldsymbol{A}$ is a sufficient condition to guarantee that every limit point of the sequence $\{(\boldsymbol{x}^q, \boldsymbol{z}^q)\}$ is an optimal solution. However, $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{I}$ must be invertible, and thus, Proof of Theorem 1 has been completed. ■

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